NUCLEAR RECEPTOR PROFILING IN HUMAN CANCERS

ABSTRACT

Nuclear receptors (NRs) are ligand activated transcription factors that play important roles in development metabolism and cancer. This study used a bioinformatics approach to predict the prognostic and regulatory importance of NRs across four cancer types: ACC (Adrenocortical Carcinoma), LGG (Low-grade Gliomas), KIRC (Kidney Renal Clear Cell Carcinoma) and LIHC (Liver Hepatocellular carcinoma). Public datasets from TCGA, GEPIA2.0, GEO2R and CistromeDB were used to analyse the survival and expression patterns, differentiually expressed genes, functional gene enrichment and transcription factor binding. Major findings include the downregulation of THRB in ACC. In LGG the over expression of AR was observed, which is associated with worst prognosis and the activation of the FOXO pathway. In KIRC, PPARG, AR and RXRA act as co-regulators for other receptors and potentially forming a master regulatory network. In LIHC, higher expression of PPARG is is involved in driving resistance to sorafenib treatment. This will provide clear and novel insights into the role of NRs in human cancers. This will in turn help in forming a base for future validation experiments.

1. INTRODUCTION

Nuclear receptors (NRs) are the largest superfamily of transcription factors, comprising 48 in humans. NRs function as ligand-activated transcription factors. NRs play important physiological functions, including growth and development, immune response, energy homeostasis and pathological processes such as cancer, diabetes, immunosuppression, and cardiovascular disease. All NRs have a common modular structure, comprising four main domains. An N-terminal regulatory domain, a central DNA binding domain (DBD), and a C-terminal Ligand binding domain (LBD), which is connected to the DBD via a flexible hinge **(Fig 1.1)**. The N-terminal domain contains the transactivation domain (TD), also known as domain AF-1, which is a non-conserved N terminal region considered the most selective part of the protein as it interacts with the nuclear factors resulting in different biological effects (1–3).

The LBD plays a crucial role in binding ligands; these ligands activate the transcription process by engaging in coactivator proteins (AF-2), enabling dimerization for firm and strong DNA binding. The DBD helps receptors bind specific DNA sequences in gene promoters, known as response elements, and transmits signals to other receptor regions. The DBD is linked to the LBD by a region called the hinge region. This region consists of a short amino acid sequence and can be phosphorylated to enhance gene activation. The P-box is responsible for binding to specific DNA sequences and found within Helix 1 of the DBD. The D-box is found in the C-terminal zinc finger in the DBD, important for the dimerization of NRs. The T box is important for heterodimeric interactions and A-box helps in the stabilization of DNA interaction (4).

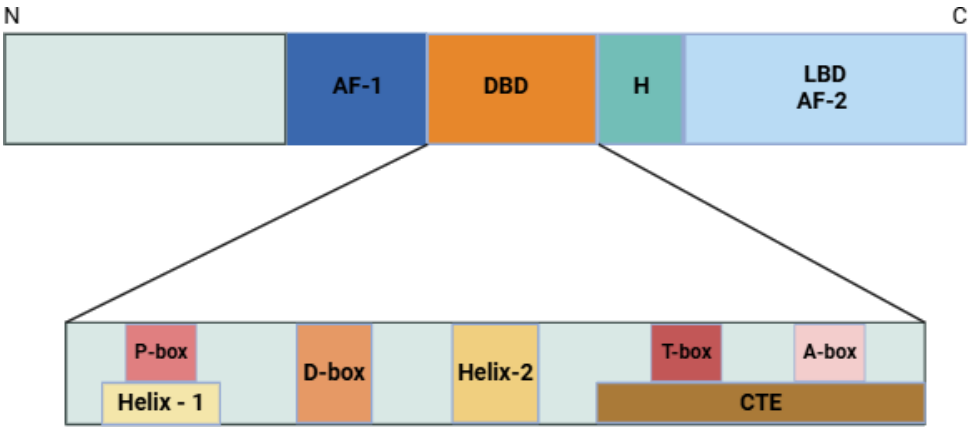


Fig 1.1: Diagrammatic representation of nuclear receptor structure. Shows the activation function (AF-1), DNA binding domain (DBD), Hinge domain (H), and Ligand binding domain (LBD). Beneath shows the more detailed view of the DBD, also representing the location of D-box, P-box, Helix-1 and 2, T-box and A-box. The P box is located within helix 1 of the DBD (Adapted from (4), using biorender)

The NR superfamily is divided into three families or classes based on their structure and mechanism of action **(Fig 1.2)**. Class I, known as the steroid receptor family, includes the progesterone receptor (PR), oestrogen receptor (ER), glucocorticoid receptor (GR), androgen receptor (AR) and mineralocorticoid receptor (MR). The Class II family, thyroid/retinoid family, have the thyroid hormone receptor (TR), Vitamin D receptor (VDR), the Retinoic acid receptor (RAR), and peroxisome proliferator activator receptor (PPAR). Class III is known as the orphan receptor family. These are a group of proteins that are identified via sequence analysis as belonging to the NR superfamily. Bile acids are a recently discovered ligand for some orphan receptors involved in certain metabolic processes such as lipid homeostasis. Even though nuclear receptors are involved in regulating gene expression, all these classes have different functions, especially in terms of their biochemical mechanisms (5). One of the important functions of nuclear receptors is to regulate transcription and they play an important role in activating or suppressing target genes by binding directly to DNA or tethering to DNA-bound transcription factors (6).

1.1. Nuclear Receptors: Classification

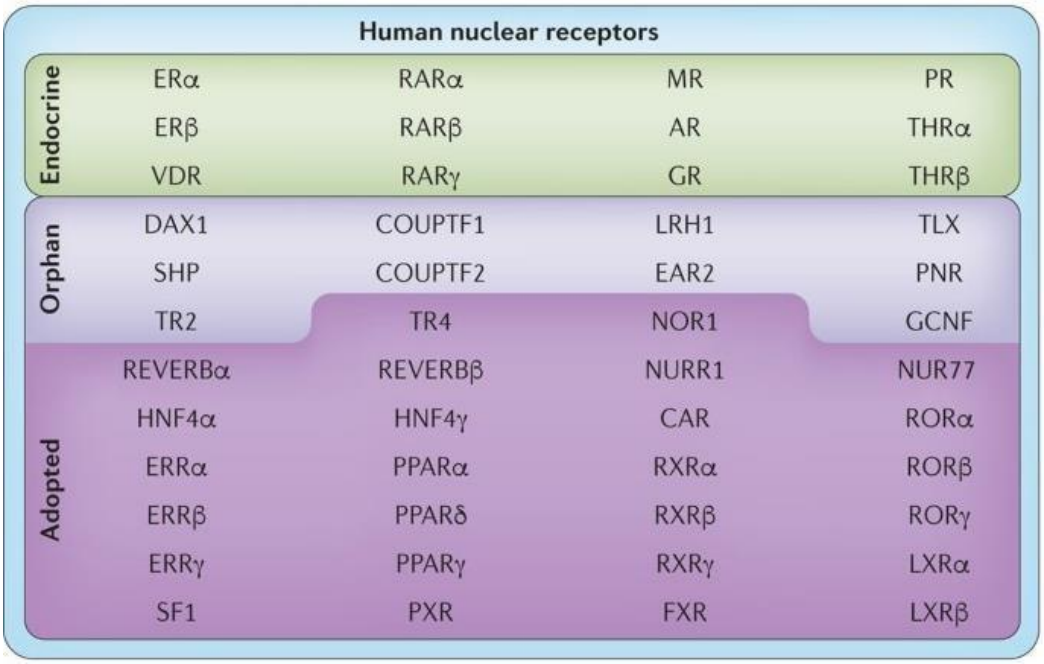


Fig 1.2: All the 48 nuclear receptors come under three subfamily categories, the endocrine, orphan and adopted receptors (Taken from (5))

The endocrine receptors are broadly classified into steroid hormone receptors and non-steroid hormonal receptors. Steroidal ligands are derived from cholesterol. The steroid hormone receptors include progesterone, oestrogen, androgens, glucocorticoids and mineralocorticoids. Oestrogen and progesterone are known as gonad steroids as they induce and maintain sexual characteristics. For many years, it has been known that steroids mediate their effects through binding and activating receptors. These receptors get activated by binding to specific hormonal ligands in target tissues and regulating the target genes' expression. All the NR superfamily has a distinct modular structure (7–9). The non-steroid receptors like Vitamin D are very important for the development of bones and calcium metabolism. Thyroid receptors, which are a different type of non-steroidal hormone receptors, are also important for development and differentiation. Their function gets induced when binding to thyroid hormones (10,11). The hormone receptor that plays a crucial role in epithelial cell growth and cellular differentiation is retinol, and its derivatives retinoic acid (12).

The next group of nuclear receptors are the orphan receptors. The significance and importance are involved in regulating metabolic pathways. Some important orphan receptors are peroxisome proliferated activated receptors (PPARs), liver X receptors (LXRs) and farnesoid X receptor (FXR). To regulate the fatty acid metabolism, the PPAR responds with the fatty acid

derivative. The main targets of PPAR are the genes that are involved in adipogenesis and lipogenesis. The LXRs function as cholesterol sensors and play a critical role in maintaining cholesterol homeostasis. So, their target genes involved in cholesterol maintain and regulate cholesterol metabolism, bile synthesis, and cholesterol absorption. FXR plays a crucial role in regulating bile acid metabolism in response to bile acid levels (13). These are subdivided into seven families, the NR0, NR1, NR2, NR4, NR5, NR6 and NR7 but NR0 do not have DBD (14).

1.2 Function of Nuclear receptors (NRs)

Nuclear receptors are ligand-regulated transcription factor which that are activated by steroid hormones and lipid soluble signals like retinoic acid, oxysterols and thyroid hormone. These ligands can cross the plasma membrane and interact with the nuclear receptor inside the cell rather than acting through cell surface receptors. Once NRs are activated, they have the capacity to regulate the transcription of various genes that control majority of the biological processes like proliferation, growth, development, metabolism and reproduction. Even though NRs function as transcription factors, studies have also found that they can regulate cellular function in cytoplasm. For example, Testosterone can act through androgen receptor in the cytoplasm to activate signaling pathways (15). Many NRs especially type II receptors like the PPARs, RARAs and TRs heterodimerize with RXRs to bind DNA effectively. Therefore, RXR is an important hub that forms heterodimers that increases the gene regulatory capabilities of receptors (16).

1.3 Nuclear Receptors in Cancer Biology

NRs have been an important focus in cancer research for decades as they are druggable targets that control pathways that are key regulators of cancer. In some cases, such as breast cancer, NRs can function as biomarkers that help in the subclassification of tumours and targets for hormone- based therapeutic approaches (5). As the most successful example of targeting an NR to treat cancer, a large amount of research has investigated NRs in breast cancer, especially the estrogen and progesterone receptors **(Fig 1.3)**. However, advances in gene expression techniques have provided limited evidence for the involvement of other ligands in breast cancer such as androgens, corticosteroids, lipids, fatty acids, and vitamins A and D. As much as breast cancer is due to estrogen receptors it is important to regulate the activity of other NRs too (17).

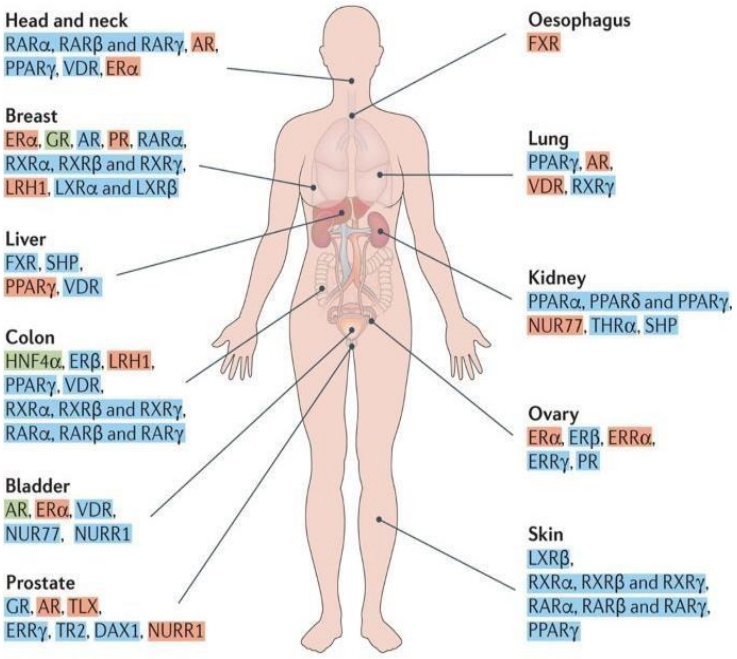


Fig 1.3: The schematic representation of different organs in our body that specifically express certain nuclear receptors. For example, in the Lungs, PPARγ, AR, VDR and RXRγ were expressed. The blue-tagged NRs are involved in preventing tumour progression. The (red) NRs are involved in tumour growth and progression. The green ones can act as both tumour suppressive and oncogenic. (Taken from (5))

Hypothesis and Aims

As discussed above nuclear receptors are transcription factors that is activated by specific hormones, lipid soluble factors and metabolites. They also tend to regulate gene expression of specific gene sets that drive key cellular processes. NRs are known as master regulators of cell fate, metabolism and immune function, and so it is not surprising that most types of cancer have altered NR activity. **Based on this, I propose that NRs are promising targets to improve patient outcomes in a range of cancers.** A detailed, unbiased analysis of NR profiles in cancers will be the first step in this process. This project will characterize the expression of NRs, and the transcriptional profiles of those NRs in a panel of cancers to identify which may be suitable candidates for experimental analysis. Specifically, I will determine:

1) Which combinations of NRs have altered expression in cancer
2) Which gene sets (and therefore pathways) those NRs control

This will indicate whether targeting a single NR, or group of NRs could be a new avenue for cancer treatment.

2. MATERIALS AND METHODS

Table 2.1: Details and IDs of all the datasets used in this project (18–28).

	ID	Study parameters	How the data was used in this study	Reference
Gene microarray	GEO ID: GSE10927	Human samples from 33 adrenocortical carcinomas, 22 adrenocortical adenomas and 10 normal adrenal cortex samples. Each sample were from different patients.	To identify genes differentially expressed in Normal adrenal cortex and Adrenocortical carcinomas (ACC).	18
	GEO ID: GSE86457	Human neural stem cell (hNSCs) treated with DHT (10 nM, 100 nM), R1881 (1nM), Retinoic acid (1μM) or Testosterone (10nM, 100nM) for 24 hours.	It Is used to identify AR target genes by comparing the ligand treatments vs the DMSO as the control	19
	GEO ID: GSE192771	Huh7 liver cancer cells which has been treated with sorafenib or DMSO	Used to identify sorafenib regulated gens in LIHC (Supports analysis of PPARG role in treatment resistance)	-
	GEO ID: GSE94550	Expression profiling of Huh7 hepatocellular carcinoma cells, untreated parental, sorafenib resistant pool and sorafenib resistant clone cells	To identify genes involved in sorafenib resistance in LIHC	20

	GEO ID: GSE125570	The HepG2 liver cancer cells were treated with Troglitazone, Pioglitazone or Vehicle i.e, DMSO for 2 and 16 hours.	Used to analyze the gene expression changes in response to PPAR ligands (which supports the involvement of PPARG in LIHC)	21
	GEO ID: GSE29159	The HepG2 cells were transfected with wildtype or mutant thyroid hormone receptors, THRA and THRB. The gene expression is analyzed using T3 treatment.	Used to study the regulation by THRB in ACC. Also helped to identify THRB regulated genes in cancer.	22
Nuclear receptor ChIP-seq	Cistrome ID: 76155	AR ChIP-sq in LNCaP prostate cancer cells that is being treated with R1881	To identify AR target genes in LGG	23
	Cistrome ID: 50043	AR ChiP-seq in LNCaP prostate cancer cells which is being treated with Dihydrotestosterone (DHT)	to compare the DHT-induced AR binding with other treatments in LGG	24
	Cistrome ID: 50100	AR ChIP-seq in LNCaP prostate cancer cells treated with Bicalutamide	To compare an antagonist (bicalutamide) against agonists (DHT, R1881) in LGG analysis.	24
Transcription factor ChIP-seq	Cistrome ID: 71216	FOXA1 ChIP-seq in LNCaP prostate cancer cells.	To identify FOXA1 binding sites for	25

			overlap with AR targets in LGG.	
	Cistrome ID: 101423	PPPARG ChIP-seq in HepG2 hepatocellular carcinoma cells.	UCSC genome track for visualizing PPARG binding at E2F8 regions and used to download the putative targets for comparison.	26
	Cistrome iD: 46242	RXRA ChIP-seq in HepG2 cells	To visualize RXRA binding near the E2F8 gene using the UCSC genome browser.	27
Co-activator ChIP-seq	Cistrome ID: 45799	PPARGC1A ChIP-seq in HepG2 cells	Used as a gene track along with RXRA and PPARG	-
Transcription factor ChIP-seq	Cistrome ID: 38804	E2F8 ChIP-seq in LoVo colorectal adenocarcinoma	To identify and download the putative targets of E2F8	28

2.1 Dataset identification and selection

Important and relevant datasets were identified through manual searches with help of certain databases like the GEO and CistromeDB using specific keywords such as THRB, ACC, cancer, RXRA, PPARG, E2F8. Studies were also selected based on experimental relevance (for example the hepatocellular carcinoma models), the cell type (HepG2) and the availability of data relevant to the study. For the transcriptomics, the GEO2R was used to collect and identify differentially expressed genes and the CistromeDB was used to collect the ChIP-seq of relevant genes and transcription factor. It is also used to download the putative target gene list and to analyze the genome browser tracks. All the thirteen databases listed in **Table 2.1** were selected and reviewed based on its quality

Table 2.2: Summary of all computational platforms and tools used in this project

PLATFORM USED	FUNCTION	URL
The Cancer Genome Atlas	It contains the raw data from thousands of normal and matched tumour tissues across 33 cancers.	https://cancer.gov/tcga
GEPIA2	It's a kind of analysis platform which hosts data from TCGA. This enables access to expression data, survival data, correlation analysis and gene list which predict patient outcome.	https://Gepia2.cancer-pku.cn
GEO2R	This tool generates differentially expressed genes from microarray and RNA-seq data in GEO accession viewer.	https://ncbi.nlm.nih.gov/gds/
CISTROME	A repository for ChIP-seq data	https://cistrome.org
UCSC Genome Browser	Extracted the gene tracks for AR, FOXA1, PPARG and its associated Transcription factors.	https://genome.ucsc.edu/
ENRICHR	A tool used to analyse the pathways, ontologies and transcription for a particular gene list.	https://maayanlab.cloud/Enrichr/
STRING	Predicted protein-protein interaction network and a tool used for functional enrichment analysis.	https://string-db.org
TF LINK	Provide comprehensive and highly accurate information's about transcription factors-target gene interactions.	https://tflink.net/

2.2 Filtering, Selection and Formatting of Data

2.2.1 Transcriptomics Data

At first, the selected dataset was processed using GEO2R **(Table 2.2).** Then the groups were correctly defined and got the differentially expressed gene lists and its adjusted p-values. All these results were exported as text files in Excel. The genes from these lists were selected based on the p-value ($p<0.05$) for further analysis. All the gene lists obtained from CistromeDB and GEO2R were already formatted as HGNC gene symbols, no further conversion was required.

2.3 Data Overlays and Analysis

All the analysis in this project was conducted using Microsoft Excel and the databases listed in **Table 2.** All bar graphs were made using Graph Pad Prism and the Venn diagrams were created using GeneVenn for comparison.

2.3.1 Data Overlays

For creating the Venn diagram, all lists were imported as gene symbols and resulting individual genes and overlapping genes were exported as plain text into excel file. Then the Venn diagrams were redrawn using Microsoft PowerPoint

2.3.2 ENRICHR

ENRICHR serves as a platform that connects the gene lists provided by the user to a wide range of public datasets and helps in the identification of functional enrichment. The valid Entrez gene symbols were pasted or exported into the home page of Enrichr. After running the page, different tabs can be analyzed based on our interest and which is relevant to the study. The pathway and the transcription tabs were used predominantly for this project. The enriched terms in these tabs were analyzed and filtered based on the adjusted p-value. Just by hovering the enriched terms, gives the genes that regulate them. The tables were exported as a text file which can be used for processing in GraphPad Prism.

2.3.3 STRING

STRING is a database that collects and build networks of already known protein-protein interaction (29). The gene symbols were used as the input in the homepage of STRING where it says ‘Multiple Proteins’ and selected Homo sapiens. After that we get the STRING diagram and identify the genes. The settings tab was used to adjust the network type, the meaning of each network edges. The analysis tab can be used to colour the nodes of specific receptors relevant to KIRC.

2.3.4 CistromeDB

The Cistrome was initially studied with a method called ChIP-ChIP that combines chromatin immunoprecipitation with microarrays. However, with help of next generation sequencing, a new method called ChIP-seq was becoming more predominate and widely used (30). So, the Cistrome database is a repository for ChIP-seq data.

CistromeDB, was searched for the transcription factor of interest and under the species tab 'Homo sapiens' was selected. The IDs of the tissues were also checked if it was available. During the analysis, the ones that had been previously published were favoured rather than selecting data that was deposited. Also, the QC reports were analysed and selected the ones where it passed all QC (all circles were green). If there is more than one option, the QC report will be analysed and check the number of peaks that has been detected, basically the ideal number of peaks should be greater than 10,000. However, the shape of the peaks also matters, whether its tall and narrow **(Fig 2.1)**. This has been used for selecting the datasets relevant to to this project. Then the putative targets were downloaded and exported as a text file in excel. The gene list was selected based on a score greater than 2 (score>2). In CistromeDB, the UCSC genome browser can be accessed that opened different tracks.

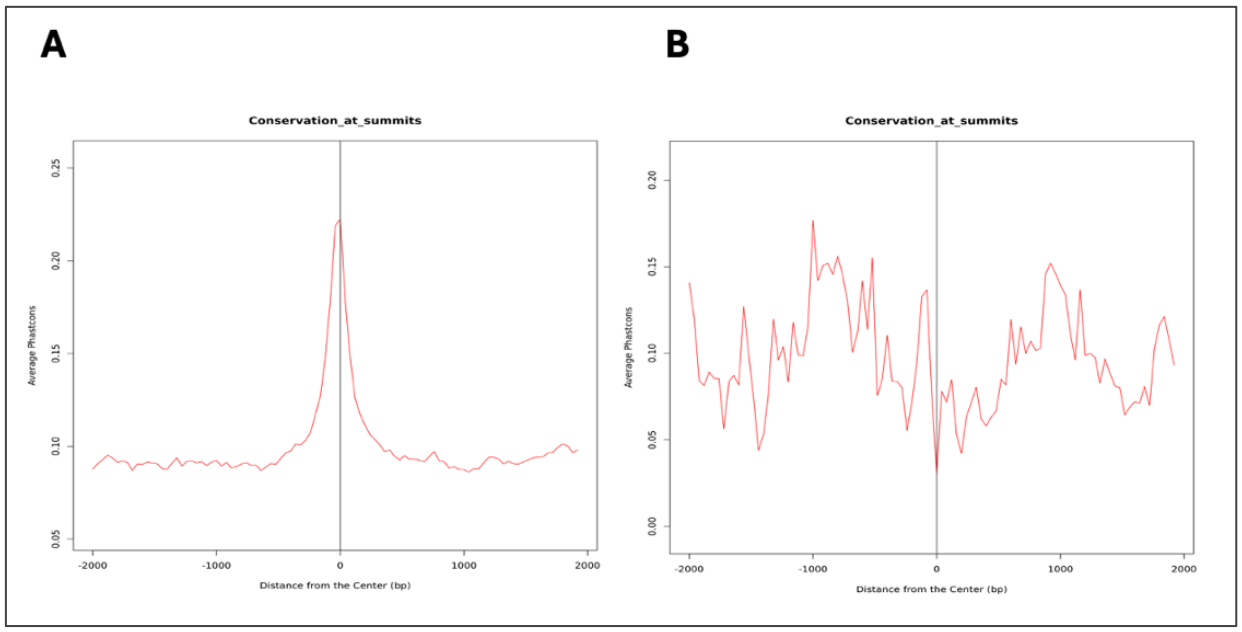


Fig 2.1: This represents how the ChIP-seq dataset was chosen for the analysis. As in the figure the shape of the peaks is narrow and tall indicating the quality of the peak (Top one)**.** The bottom plot shows a low-quality peak. The shape of the peak was analysed for selecting ChIP-seq dataset.

In UCSC genome browser, the E2F8 transcription factor was searched to get the gene track. This result was taken as a screenshot.

3.RESULTS

To begin this project, we first aimed to explore the potential role of nuclear receptors (NRs) in different cancers, without focusing and restricting ourselves with any one receptor or cancer type. So, our goal was to identify specific nuclear receptor and cancer pairs that biologically and clinically relevant based on the relation with patient survival. To achieve this, we first conducted survival analysis using GEPIA 2.0 platform, which provides the Kaplan-Meier curve (survival curves) for gene expression data across TCGA datasets. This helped us to analyze and study how different nuclear receptors were associated with overall survival (OS) in different types of cancers. After careful analysis, I picked cancers that showed interesting patterns. So, I initially selected 14 cancers that showed significant survival associations with different nuclear receptors These cancers are ACC, BLCA, STAD, PAAD, KIRC, LGG, LIHC, MESO, SKCM, UVM, READ, LUSC, BRCA, LUAD **(Table 3.1)**.

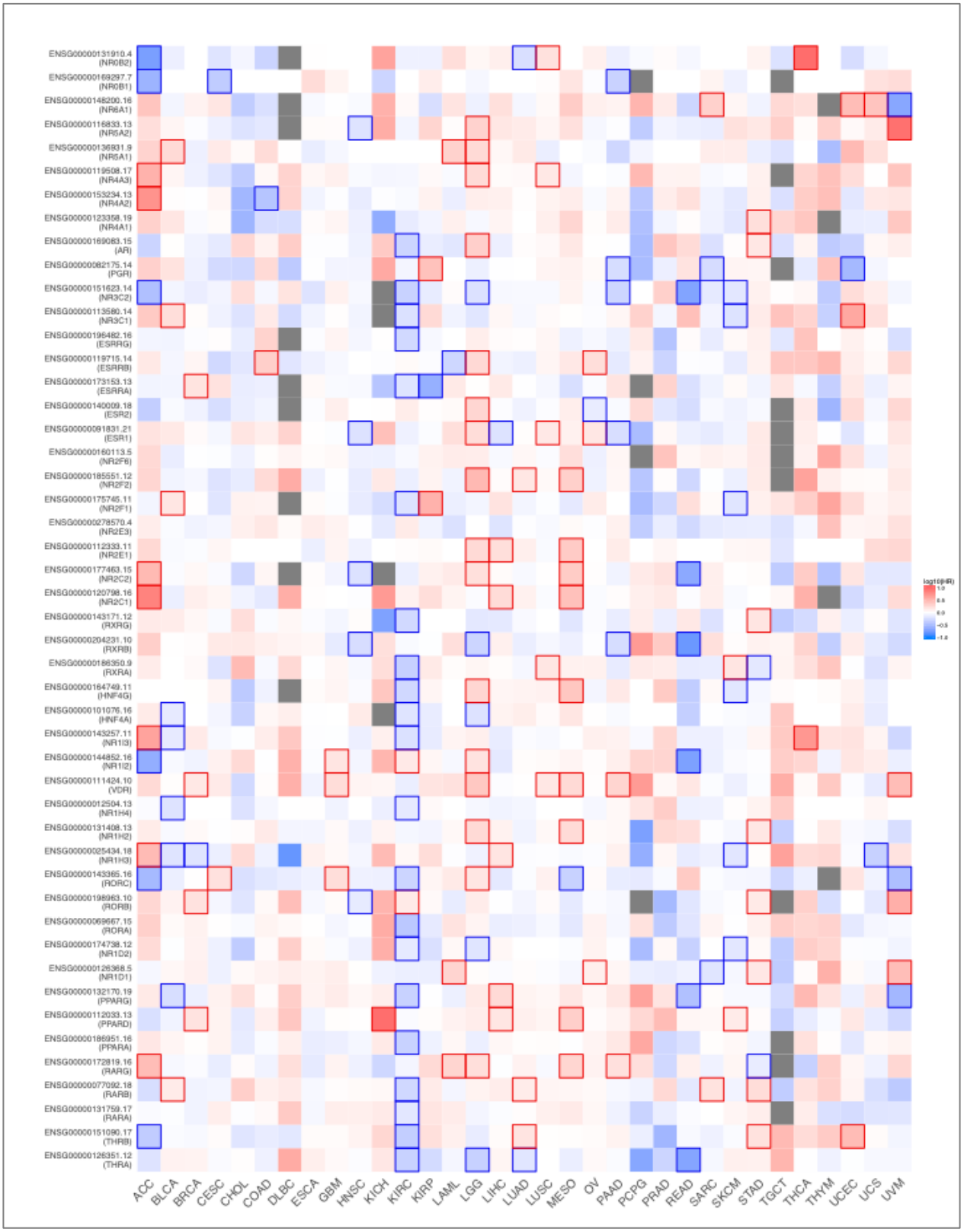


Fig 3.1: The survival map (overall survival (OS)) of NRs across different cancer types: This map demonstrates the survival association between the gene expression of 48 nuclear receptors across 33 cancer types which was made using the GEPIA2.0 platform. Each row corresponds to a nuclear receptor and each column to a cancer type. The intensity of the colour demonstrates the value of the Hazard ratio (HR), the red indicates HR>1 where high expression associated with worst prognosis and the blue indicates HR<1, plays a protective role. And the statistically significant ones

were highlighted with red and blue borders (p≤0.05). The survival map helped in the selection of all the 14 cancer types for the initial analysis.

Table 3.1: Details of the nuclear receptors that showed significant survival associations in various cancer types on the GEPIA2.0 analysis. For each of the selected cancer, the full name and all the associated NRs were listed.

CANCER TYPE	FULL NAME	NUCLEAR RECEPTORS
ACC	Adrenocortical carcinoma	THRB, RARG, RORC, NR1H3, NR1I2, NR1I3, NR2C1, NR2C2, NR3C2, NR4A2, NR4A3, NR0B1, NR0B2
BLCA	Bladder cancer	RARB, PPARG, NR1H3, NR1H4, NR1I3. HNF4A, NR2F1, NR3C1, NR5A1
KIRC	Kidney renal clear cell carcinoma	THRA, THRB, RARA, RARB, PPARA, PPARG, NR1D2, RORA, RORB, RORC, NR1H4, NR1I2, NR1I3, HNF4A, HNF4G, RXRA, RXRG, NR2F1, ESRRA, ESRRG, NR3C1, NR3C2, AR
LGG	Low-grade gliomas	THRA, RARG, NR1D2, RORC, NR1H2, VDR, NR1I2, HNF4A, HNF4G, RXRB, NR2C2, NR2E1, NR2F2, ESR1, ESR2, ESRRB, NR3C2, AR, NR4A3, NR5A1, NR5A2,
LIHC	Liver hepatocellular carcinoma	PPARD, PPARG, NR1H3, NR2C1, NR2E1, ESR1,
MESO	Mesothelioma	RARG, PPARD, RORC, NR1H2, VDR, HNF4G, NR2C1, NR2C2, NR2E1, NR2F2,
SKCM	Skin cutaneous melanoma	PPARD, NR1D2, NR1H3, HNF4G, RXRA, NR2F1, NR3C1, NR3C2,
STAD	Stomach adenocarcinoma	THRB, RARB, RARG, NR1D1, RORB, NR1H2, RXRA, RXRG, AR, NR4A1
UVM	Uveal melanoma	PPARG, NR1D1, RORB, RORC, VDR, NR5A2, NR6A1

READ	Rectum adenocarcinoma	THRA, PPARG, NR1I2, RXRB, NR2C2, NR3C2,
LUSC	Lung squamous cell carcinoma	VDR, RXRA, ESR1, NR4A3, NR0B2
PAAD	Pancreatic adenocarcinoma	RARG, VDR, RXRB, ESR1, NR3C2, PGR, NR0B1
LUAD	Lung adenocarcinoma	THRA, THRB, RARB, NR2F2, NR0B2
BRCA	Breast cancer	PPARD, RORB, NR1H3, VDR, ESRRA,

After the initial screening, a table was created for each of these cancers with all the associated nuclear receptors, hazard ratios (HR), P value and their number of samples. I downloaded and analyzed the KM plots for all the nuclear receptors associated with these 14 cancers and conducted a literature study on prognosis, incidence and current treatment regimens **(Table 3.3)**. To make sure that the analysis was statistically robust, I filtered the receptors by selecting those with statistically significant survival associations i.e., p-value ≤0.05. those receptors having a HR>2 was associated with poor prognosis and those with HR<0.5 play potential protective roles.

Table 3.2: Flagging of cancer types for NR analysis. Green- high significance, yellow- moderate significance and red-no significance

Type of Cancer	Flag colour	Significance
ACC	Green	Strong survival association and expression pattern
LGG	Green	Strong survival association and expression pattern
KIRC	Green	Strong survival association and expression pattern
LIHC	Yellow	Moderate significance
STAD	Yellow	Moderate significance

MESO	Yellow	Moderate significance
PAAD	Yellow	Moderate significance
Others	Red	No significance

So, to prioritize which cancers and receptors to choose for further analysis, we did a flagging system (Green, Red and Yellow). If its green then: very high priority (statistically strong, poor prognosis and incidence), yellow: potential back up option and red: no significance, not going to choose for further analysis **(Table 3.2)**. So, we done a thorough and in-depth analysis to choose the cancer for downstream analysis and selected some cancer types with strong survival and expression patterns. Those cancer are ACC, LGG and KIRC were flagged green. LIHC, STAD, MESO AND PAAD were flagged yellow due to moderate statistical significance. To explore and validate these findings, I performed the box plot and stage plot analysis for each of the receptors associated with each of these cancers. This helped us to understand which how the expression of each receptor changes across tumour stages and subtypes. I selected those receptors that showed consistent high or low expression across box and stage plot. So, we did the final short listing based on consistent survival association and expression patterns, biological interest and research potential especially in cancer **(Table 4.1, 5.1, 6.1 and 7.1)**.

Table 3.3: Shows the details of the selected four nuclear receptors and its associated cancers based on its prognosis, incidence and current treatment strategies.

CANCER TYPE	INCIDENCE	PROGNOSIS	CURRENT TREATMENTS
ACC	0.7-2.0 per million	10%-60% (5-year survival)	Surgery, radiotherapy and chemotherapy
LGG	4.7 per 100,000 person-years	Type 1 gliomas: 5-year survival, 95% Type 2 gliomas: 5-year survival, 40%-50% In general: 4.7-9.8 years	Surgery, radiotherapy, chemotherapy
KIRC	400,00 new cases annually	50%-69% (5-year survival)	Surgery, radiotherapy, chemotherapy, immunotherapy and targeted therapy

LIHC	7.3 cases per 100,000 people globally	21%, 5-year survival	Surgery, radiotherapy, chemotherapy, immunotherapy, targeted therapy and liver transplant

With the help of this approach, we narrowed our focus down to four cancer types and four respective nuclear receptors and those are: ACC - THRB: with the drug treatment & normal vs cancer LGG - AR: FOXA1 & AR on different genes, LIHC- PPARG: focusing on drug resistance, KIRC- I am going to look at which NR are regulated by each other. This was clearly demonstrated in **Table 3.4.** Therefore, each of this selected pairs were analyzed and explored in detail in the following sections.

Table 3.4: Details of all the selected pairs for further analysis. The 'X' mark indicates the selection for analysis. LOW- low expression and HIGH- high expression. This table shows the receptors that showed consistent expression in both stage and box plots

Nuclear receptor	**ACC**	**LGG**	**KIRC**	**LIHC**
THRB	X (LOW)		X (LOW)	
THRA		X	X (LOW)	
RARB			X (LOW)	
PPARA			X (LOW)	
PPARG			X (LOW)	X (HIGH)
NR1D2		X	X (LOW)	
RORA			X (LOW)	
RORC	X (LOW)	X	X (LOW)	
NR1I3	X		X (LOW)	

RXRA			X (LOW)	
RXRG			X (LOW)	
NR2F1			X (LOW)	
ESRRG			X (LOW)	
NR3C2	X (LOW)	X	X (LOW)	
AR		X (HIGH)	X (LOW)	

So, this is all the possible nuclear receptors and all possible cancers. However, I decided on focusing these cancers on each of the remaining four chapters. Therefore, each of this selected pairs were analyzed and explored in detail in the following sections.

4. ADRENOCORTICAL CARCINOMA (ACC)

Table 4.1: Summarizes the survival and expression analysis of all the receptors associated with ACC. It includes the number of samples, the Hazard ratio, p-value and the expression of different nuclear receptors across stage plot and box plot.

Nuclear receptor	SAMPLE NUMBER	Hazard ratio	P value	Worst prognosis (HR for OS)	Subtype Expression	Stage plot F (P)
THRB	76	0.42	0.031	0.42 (LOW)	*HIGH, *INTER (down)	
RORC	75	0.27	0.0012	0.27 (low)	*HIGH, *INTER (down)	3.12 (0.0314) (low)
RARG	76	2.8	0.0087	2.8 (high)	*LOW (down)	
NR4A3	76	6.5	0.00011	6.5 (high)	*LOW (down), *HIGH (up)	2.99 (0.0366) (high)

NR4A2	76	2.6	0.016	2.6 (high)	*HIGH, *INTER, *LOW (down)	
NR3C2	76	0.45	0.04	0.45 (low)	*HIGH (down)	
NR2C1	76	4.3	0.00034	4.3 (high)		5.76 (0.00138) (high)
NR1I3	75	2.8	0.01	2.8 (high)		
NR1I2	76	0.26	0.0053	0.26 (low)		
NR1H3	76	2.9	0.0088	2.9 (high)	*HIGH, *INTER, *LOW (down)	
NR0B2	76	0.23	0.00026	0.23 (low)	*HIGH, *INTER (down)	

NOTE:" WORST PROGNOSIS (OS)" indicates the hazard ratio values from survival with relative expression that is High or Low

The results in table 4.1 shows that there were many potential nuclear receptors associated with Adrenocortical carcinoma (ACC), but I found that THRB was the most interesting among them. THRB stood out due to its strong association with poor prognosis where HR=0.42 nad p-value= 0.03, also showed consistent low expression across different subtypes and stages. So here is the Box plot and Stage plot for THRB (**Fig 4.1)**

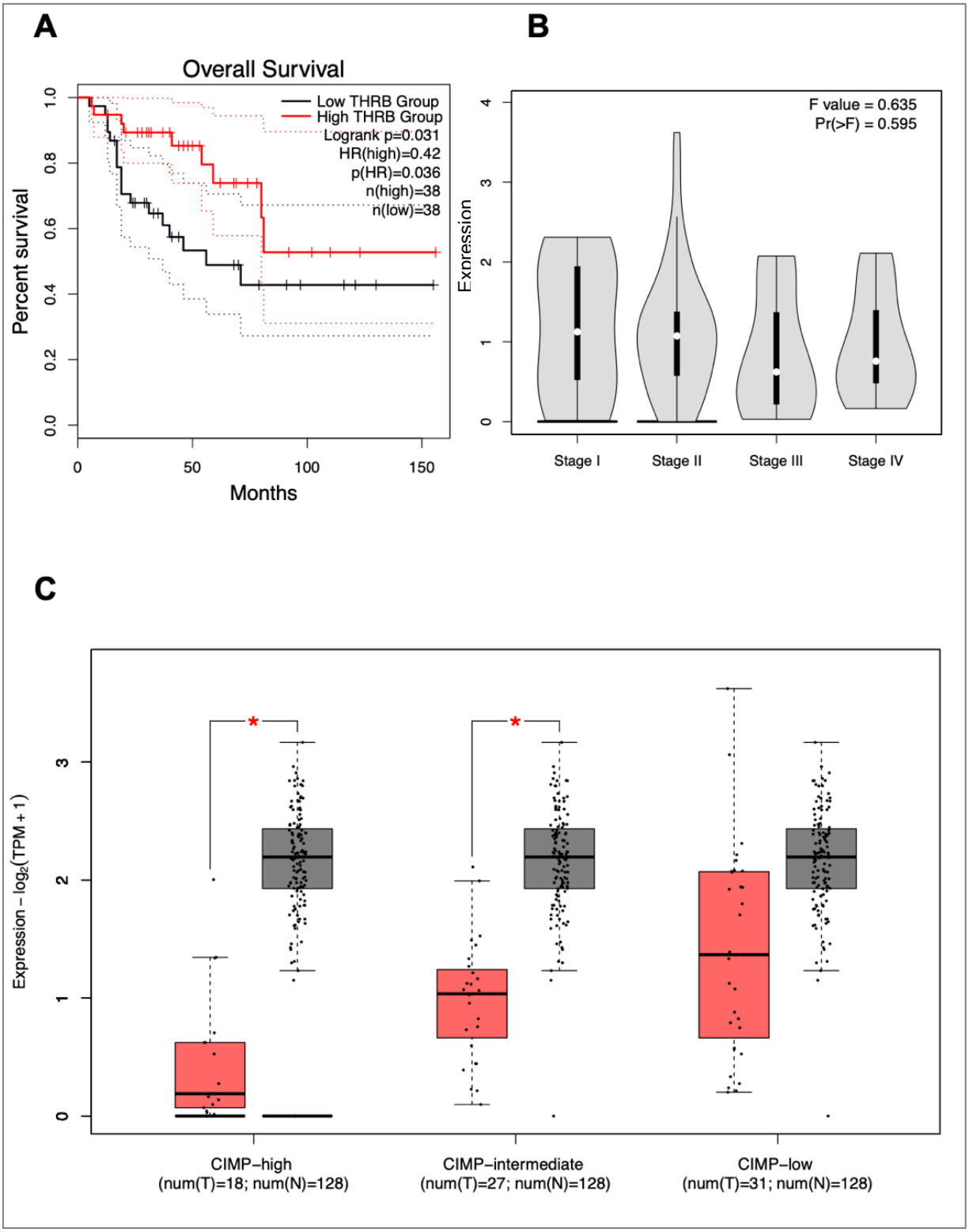


Fig 4.1. Expression and Prognostic significance of THRB in Adrenocortical Carcinoma: (A) The Kaplan-Meier survival curve which compares the overall survival between ACC patients with high and low expressions of THRB. The high expression of THRB is associated with improved survival (HR=0.42 and p-value= 0.036). This clearly indicates that THRB plays tumour suppressive roles in ACC. **(B)** Shows the stage plot of the THRB expression in four different stages (I-IV) and the Y-axis indicates the relative expression levels, This plot shows no statistically significant differences where the F value= 0.635 and F(P)= 0.595. However,, this demonstrates consistent

expression throughout the progression even though the p-value is relatively high. **(C)** The Box plot which compares the THRB expression across different subtypes (CIMP-high, CIMP-intermediate and CIMP-low).in the figure it's clear that the expression of THRB is significantly reduced in the subtypes CIMP-high and CIMP-intermediate (red) compared to the normal tissues (grey). The red star in the diagram represents its significance.

Out of all receptors associated with ACC, THRB showed statistically significant survival association with hazard ratio of 0.42 which proves that it plays a protective effect in patients. The expression pattern for THRB is also lower across different tumour types and stages based on the box plot and stage plot. So, this indicates that THRB might act as a tumour suppressor in ACC. If that's true that means that THRB is regulating genes that are important in ACC. So, I downloaded the gene lists (ACC vs normal using the dataset from GEO2R) to identify the genes whose expression is changed in ACC. Then, analyzed which of those genes I can prove that THRB can regulate directly. So, I compared the genes differentially expressed in ACC (ACC vs normal) with THRB ChIP-seq data from CistromeDB. The Venn analysis showed 204 overlapping genes. Therefore, it says half of the genes regulated by THRB are also altered in ACC as shown in **Fig 4.2.**

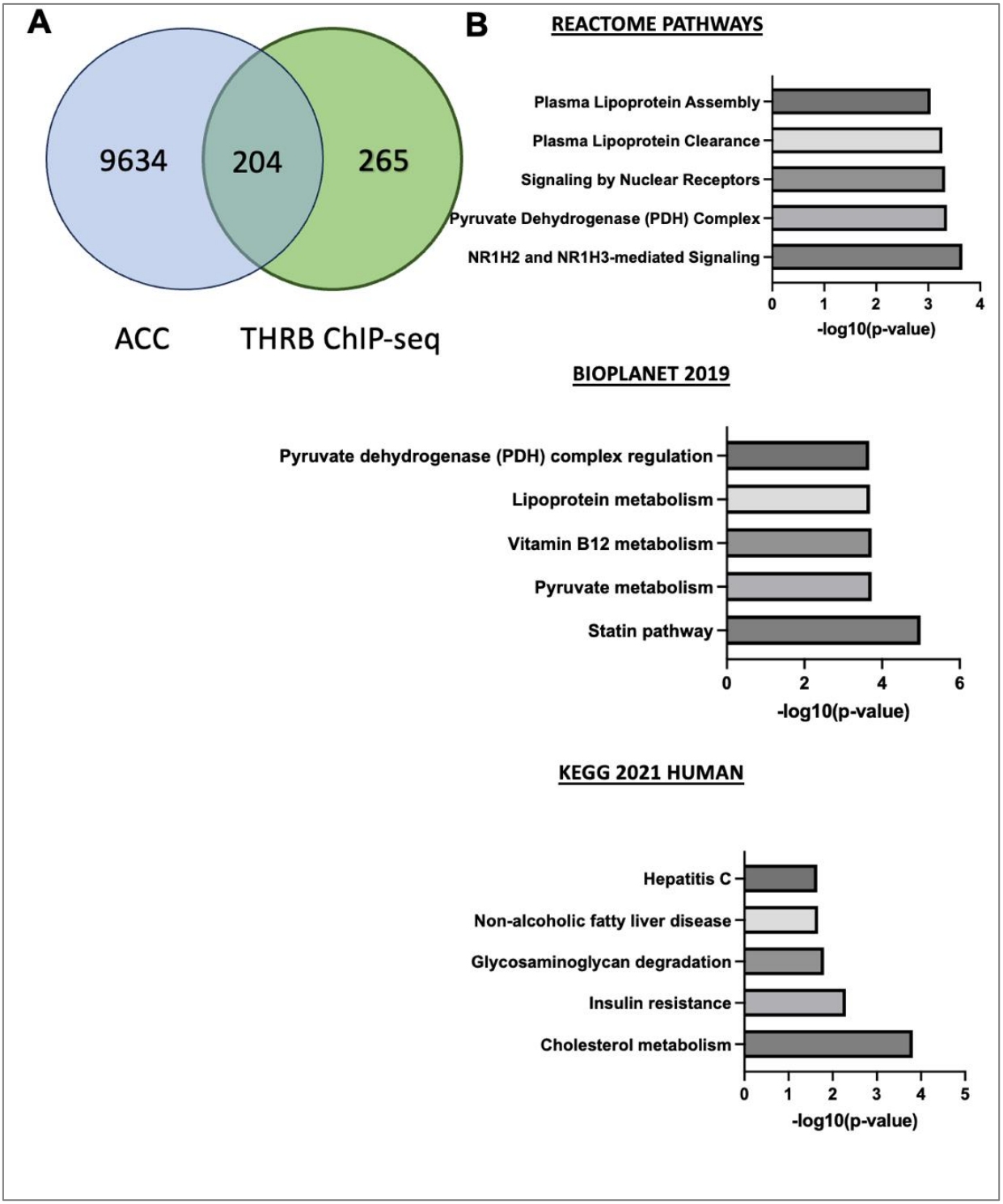


Figure 4.2: (A) This Venn diagram demonstrates the overlap of differentially expressed genes in ACC vs Normal and THRB ChIP-seq. n=9634, shows the differentially expressed genes in ACC vs Normal and the THRB bound genes (n=265) with an overlap of 204 genes. **(B)** The overlapping 204 genes were subjected to pathway enrichment analysis using ENRICHR platform via three pathways Reactome pathways, Bioplanet 2019 and KEGG 2021 Human databases. The respective bar graphs represent the biological processes regulated by the overlapping genes i.e. the top five enriched pathways. This plots clearly shows that it is involved in lipid metabolism, nuclear receptor signalling, cholesterol and insulin metabolism.

I then performed the functional enrichment for these overlapping genes using the ENRICHR. I analyzed the pathways especially Reactome pathways. Bioplanet 2019 and KEGG 2021 human and found that these genes are significantly involved in lipid metabolism and nuclear signaling pathways in which both are important to ACC. These results were then made into a bar graph to show what these genes regulate as shown in **Fig 4.2 (B).**

The first thing was to look at genes that change between the normal and cancer (ACC). Then to narrow it down, I focused on 204 genes that is known to be the THRB targets. To make this analysis more accurate, I compared these 204 genes with the Top 500 genes associated with patient survival in ACC which is based on survival analysis. The comparison was done and found 17 overlapping genes that are directly bound by the THRB and found in the top 500 genes related to survival in ACC patients. Further ENRICHR analysis was done for these overlapping 17 genes and found that these genes regulate mitotic progression, cell proliferation, DNA damage response and so on **(Fig 4.3)**. These are the key processes related to cancer.

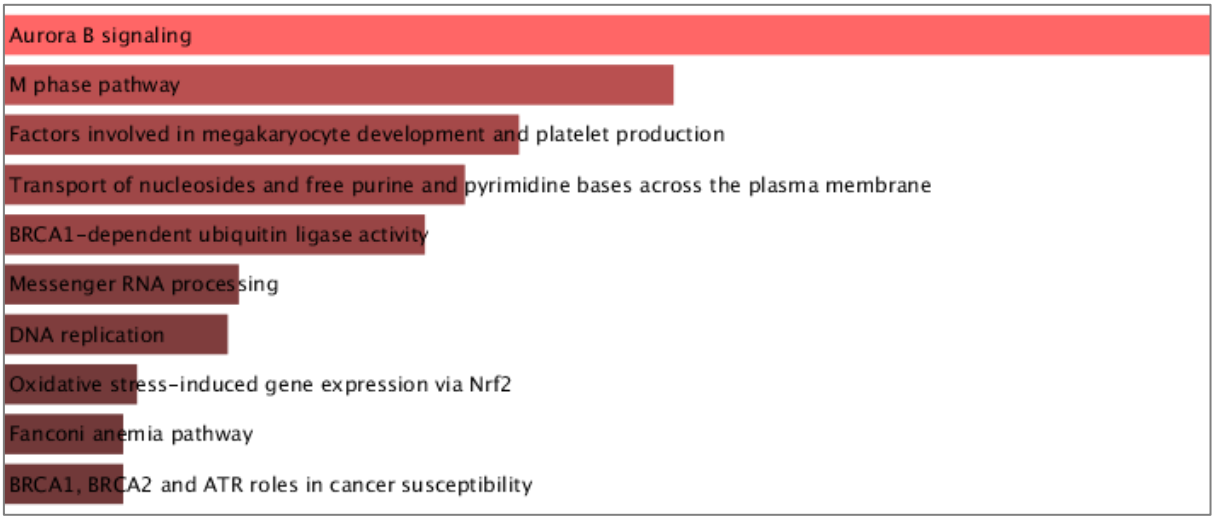


Figure 4.3: The pathway enrichment analysis of the 17 overlapping genes between THRB targets and genes associated with patient survival in ACC. The key enriched pathways are highlighted in light red.

So, to determine how THRB might influence the expression of these genes, correlation analysis was done for the 17 overlapping genes under top 500 ACC and THRB ChIP-seq. So, if it's a positive correlation, we could say that more thyroid hormone receptors are there, the more of that target gene is there and that would suggest that if you treat with Triiodothyronine (T3, which is an important thyroid hormone), it will increase the expression of that gene. If it's a negative correlation, you would predict that adding T3 would down regulate that gene. Adding T3 would improve the repression. This was done to check which genes may be activating or repressing in ACC and what effect T3 would have. Like "what role THRB playing in ACC"- oncogenic or tumour suppressive. Now, its important to check whether THRB upregulates or downregulates the expression of the 17 genes by comparing with EMPTY VECTOR T3. For selecting the best correlation graph, the p value and R value were analyzed. If R=0, No correlation and if R=1, perfect positive correlation. So, I

chose some of the genes that demonstrated perfect positive correlation and a perfect p-value as shown in the **Fig 4.4.** One such gene was CLK2 which showed a positive correlation with THRB expression (R= 0.43 AND P-VALUE = 7.7e-11). This suggests that THRB might directly target CLK2 and contributes to its tumour suppressive role in ACC

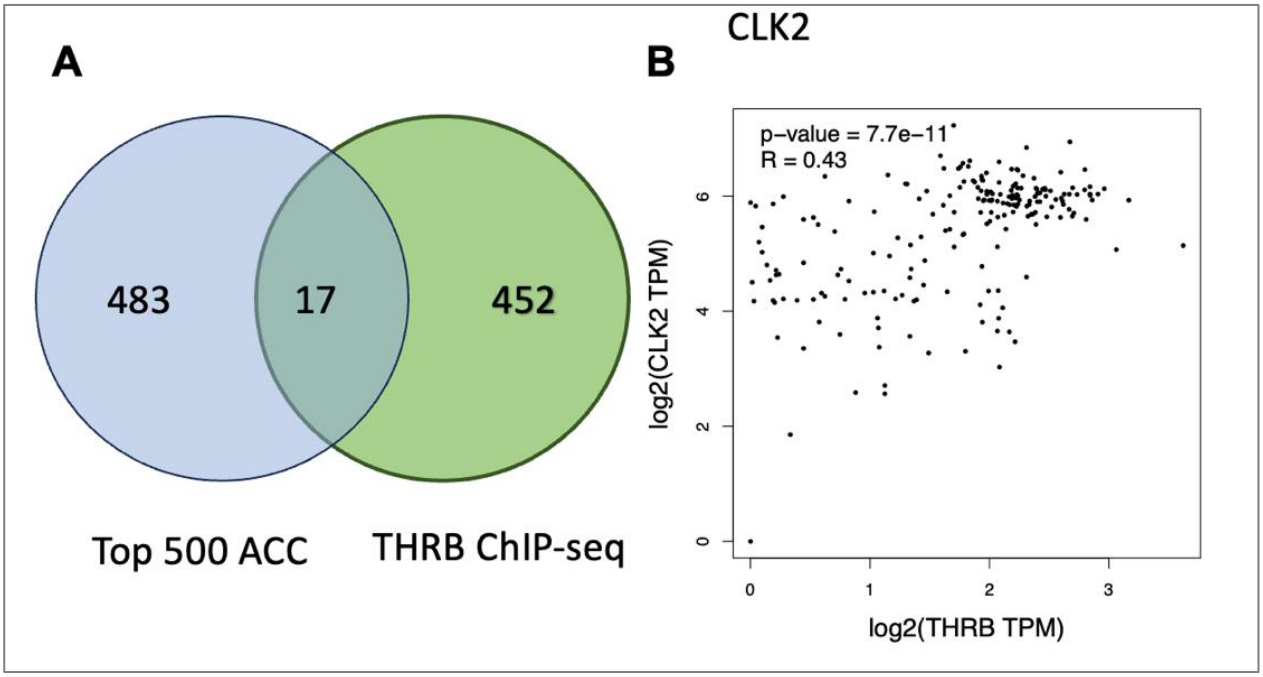


Fig 4.4: TOP 500 ACC & THRB CHIPSEQ and the correlation graph of the gene with perfect R-value. (A) The Venn diagram showing the comparison between the Top 500 ACC genes involved in patient survival and the THRB ChIP-seq, where it gives 17 overlapping genes that is being regulated by THRB. **(B).** The correlation plot of CLK2, one among the 17 overlapping genes that showed a positive correlation with THRB expression where p-value= 7.7e-11 and R=0.43. this indicates that THRB is directly involved in regulating the expression of CLK2, contributes to tumour suppressive role in ACC.

From the plot we can conclude that if the expression of THRB is higher, the expression of these genes will be also higher. Since these 17 genes are associated with survival in ACC, this indicates that THRB are involved in activating genes that suppress tumour growth and promote cell regulation.

All these together proves that THRB act as a tumour suppressor or repressor in ACC pointing that they play a protective regulatory role. This further makes THRB as a potential and strong candidate for future study as a biomarker or therapeutic target in ACC

5. LOW-GRADE GLIOMAS (LGG)

Table 5.1: Summarizes the survival and expression analysis of all the receptors associated with ACC. It includes the number of samples, the Hazard ratio, p-value and the expression of different nuclear receptors across stage plot and box plot.

Nuclear receptor	SAMPLE NUMBER	Hazard ratio	P value	Worst prognosis	Subtype expression
VDR	494	2.4	4.6e-06	2.4	
THRA	514	0.46	1.9e-05	0.46	
RXRB	514	0.63	0.011	0.63	
RORC	476	1.5	0.031	1.5	
RARG	514	1.7	0.0046	1.7	
NR5A2	502	1.6	0.014	1.6	
NR3C2	514	0.67	0.027	0.67	
NR2F2	512	2.4	2e-06	2.4	
NR2E1	514	2.2	1.3e-05	2.2 (high)	*Astrocytoma, *Oligoastrocytoma (down)
NR1H2	514	1.8	0.001	1.8	
NR1D2	514	0.64	0.014	0.64	
HNF4G	510	1.9	4e-04	1.9	
HNF4A	403	0.62	0.018	0.62	
ESRRB	514	1.5	0.024	1.5	
ESR2	488	1.6	0.011	1.6	
ESR1	495	1.5	0.023	1.5	
AR	512	2.2	3.9e-05	2.2 (high)	*Astrocytoma, *Oligoastrocytoma (up)

Same as ACC, Low grade gliomas (LGG) also have potential nuclear receptors to look at, but I found AR was the most interesting. Among all the nuclear receptors associated with LGG, AR (androgen receptor) stood out as it showed significant association with patient survival. It had a Hazard ratio of 2.2, which indicates that higher expression of AR is associated with worst prognosis. Also, the box plot and stage plot of AR showed consistent expression patterns as shown in **Fig 5.1,** which justifies the rea son for the selection of this NR.

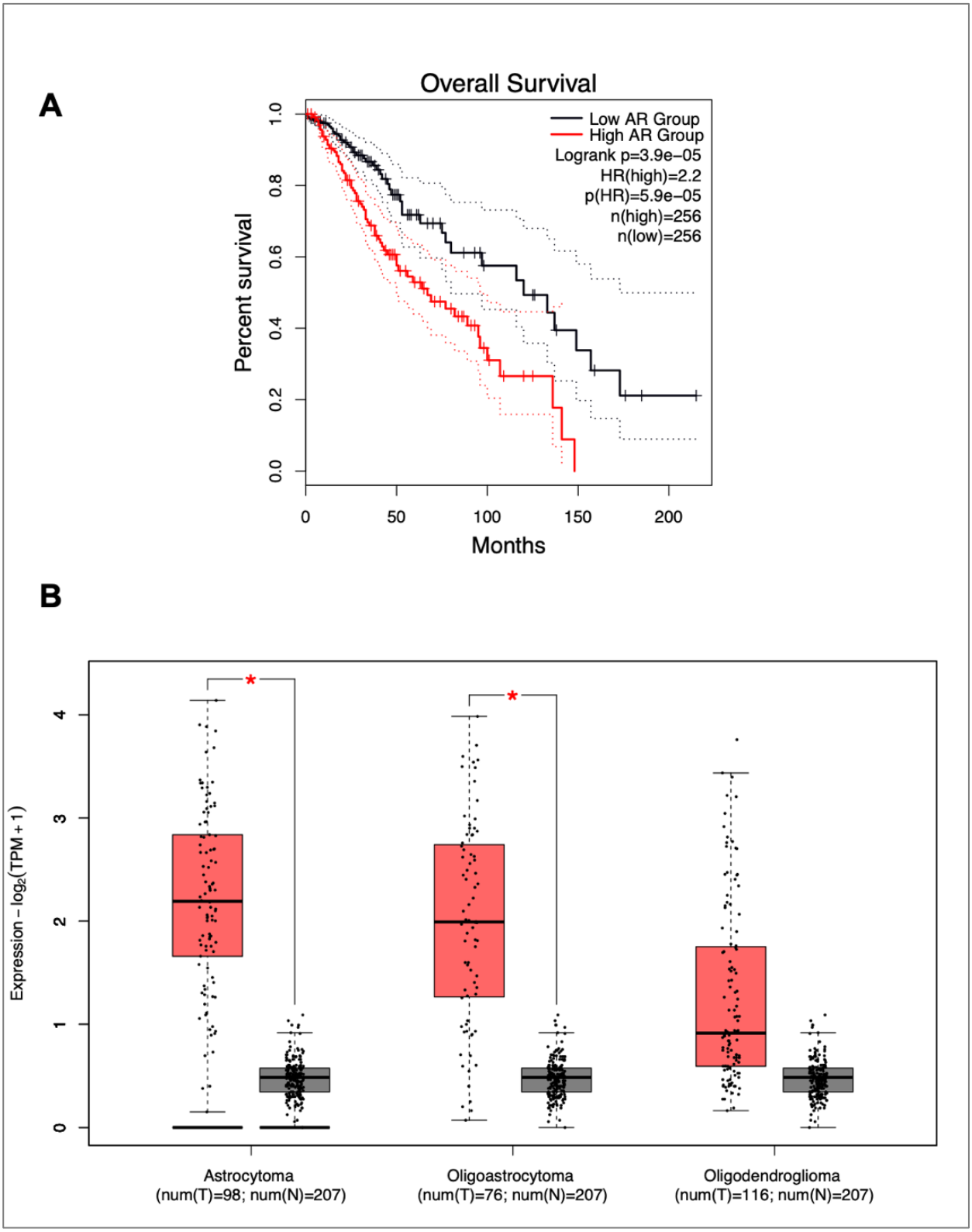


Fig 5.1: The significance and subtype expression of AR in LGG. (A) The Kaplan-Meier survival curve which compares the overall survival between LGG patients with high and low expressions of AR. As shown in the figure the high expression of AR is associated with worst survival and low the expression shows that it plays a potential protective role in LGG. **(B)** The box plot of AR across three subtypes of LGG and i.e. astrocytoma, oligoastrocytoma and oligodendroglioma. The expression of AR is significantly ($p<0.05$) higher in astrocytoma and oligoastrocytoma compared to

the normal tissues. This suggests that the higher expression of AR can be seen in these glioma forms

To further investigate the regulatory role of AR in LGG, I started by identifying differentially expressed genes from RNA-seq data obtained from GEO2R using three separate androgenic treatments i.e. DHT (Dihydrotestosterone), TESTO (testosterone) and R1881 each of these were compared against DMSO treated control. All three ligands are androgen receptor agonists that bind and activates androgen receptor signalling. Differential expression analysis was performed for each ligand versus DMSO and grouped them as DHT vs DMSO, TESTO vs DMSO and R1881 vs DMSO. Just by comparing the treated vs untreated we can identify genes that are altered due to the activation of AR. So, I got the gene list from GEO2R and compared all these lists using VENN diagram as shown in **Fig 5.2 (A).** Next, I collected the best AR ChIP-seq (R1881 treated LNCaP cells) data from CistromeDB and those are R1881 treated LNCaP, LNCAP_DHT and BICALUTAMIDE LNCAP that showed relatively high-quality peaks. But I had to select the best among these three. This was finalized after doing a comparison using Venn diagram and found that R1881 treated LNCaP has the highest number of differentially expressed genes (1220 genes) **(Fig 5.2 (B)).** This dataset was used for further analysis.

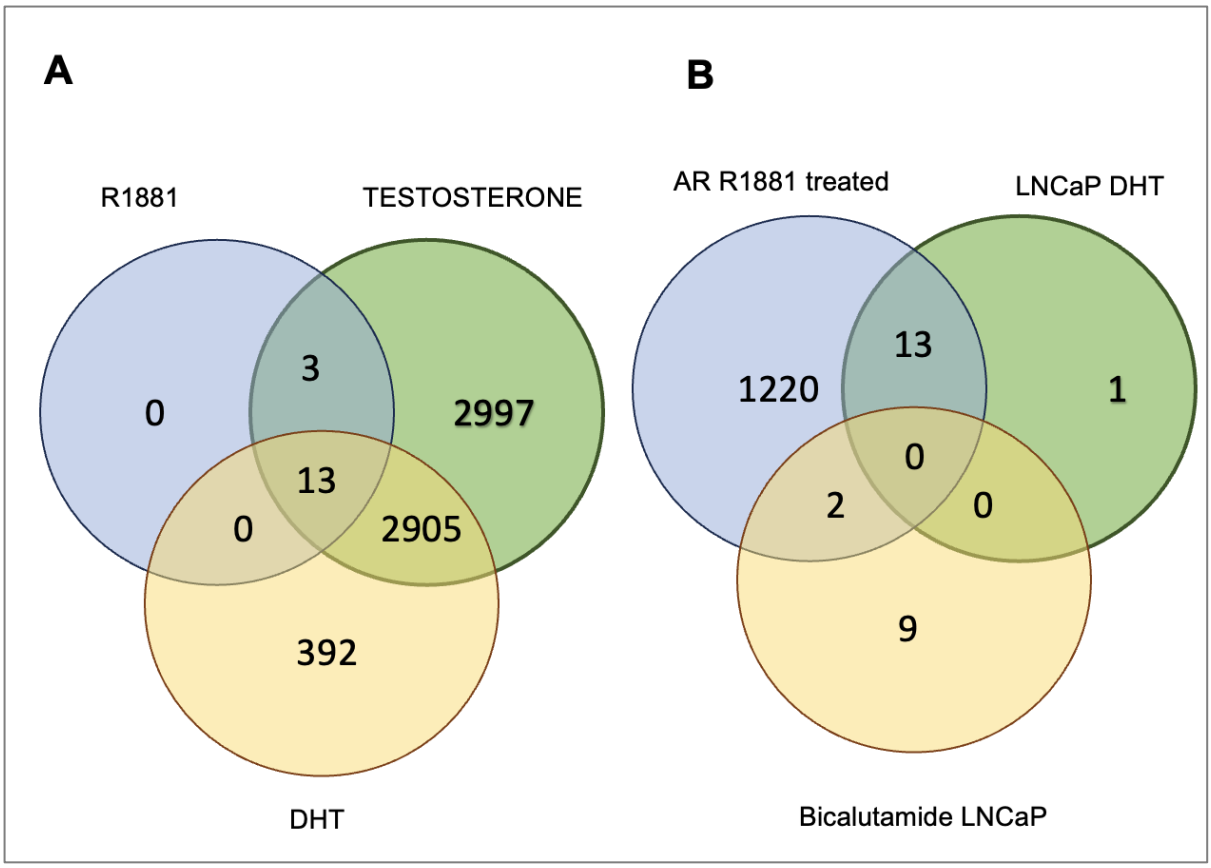


Fig 5.2: (A) The first Venn diagram shows the overlap of differentially expressed genes after treating with R1881, Testosterone and DHT in brain cell models. 13 overlapping genes were

common in all these three androgen ligands. **(B)** Comparing AR regulated genes from the ChIP-seq dataset in LNCaP cells that is being treated with R1881, DHT and Bicalutamide (AR antagonist). It's clearly demonstrated in the figure that the R1881 treated AR yielded the highest number of genes (1220).

Another comparison was done with the RNA-seq (R1881 vs DMSO) and AR Chip-seq datasets (R1881 treated LNCap cells) using Venn diagrams. This resulted in 195 overlapping genes (**Fig 5.3)**, which indicates that they might be direct AR targets. To further understand the biological functions of these genes and what they regulated, I did the functional enrichment using ENRICHR with these overlapping genes. I found very strong enrichment in the FOXO signaling pathway which tells us that AR may regulate FOXO activity in LGG. I also wanted to figure out the AR targets so I I used the top 500 LGG genes that predicts the patient survival. Then I compared with R1881 ChIP-seq gene list (to know the clinical relevance) as shown in **Fig 9.**

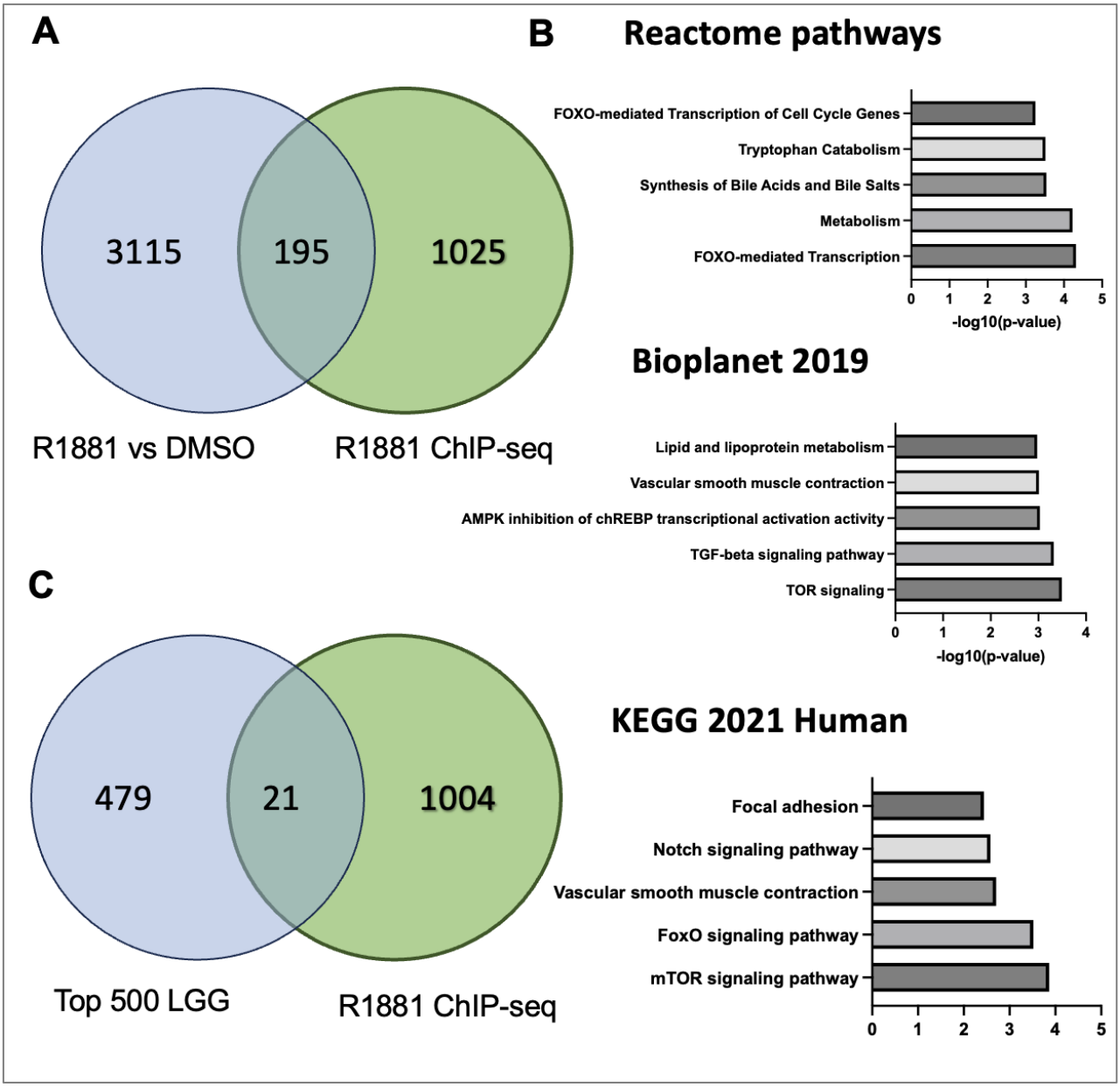


Fig 5.3: RNA SEQ vs R1881 (VENN OVERLAP) & the associated bar graph showing the 195 overlapping genes. (A) The Venn diagram showing the overlap of differentially expressed genes between R1881 vs DMSO and AR bound genes from the R1881 ChIP-seq data. A total of 195 overlapping genes were found in both RNA-seq and directly bound by AR (R1881 ChIP-seq). **(B)** The functional enrichment analysis of the 185 genes was done using the Reactome Pathways, Bioplanet 2019 and KEGG 2021 Human. Lipid metabolism, FOXO mediated transcription, mTOR and NOTCH signaling were enrich. This indicates the role of AR in diverse cancer pathways. **(C)** The venn diagram showing the Top 500 survival associated genes in LGG and the AR ChIP-seq targets, resulting in 21 overlapping genes directly regulated by AR.

Another interesting thing that stood out is that while going through the CHEA 2022 of the 195 overlapping genes under the transcription section in the ENRICHR, AR and FOXA1 ChIP-seq VCaP datasets were both enriched. So, the next thing I did was to compare the AR and FOXA1 target gene

lists. The gene list was collected from the ENRICHR. After comparison I found 57 overlapping genes **(Fig 5.4).** So, I did the functional analysis of AR and FOXO genes. I ran the ENRICHR analysis on the 57 overlapping genes. I found that these 57 genes regulate Rho GTPase signaling, it was more prominent and highest ranked than other pathways in the enrichment analysis. This pathway is associated with tumour cell motility and invasion. Lastly, I performed Chip binding analyses of AR and FOXO3 for 6 key Rho GTPase genes using UCSC gene browser by setting 2 different tracks 1) AR ChIP-seq track 2) FOXO1 ChIP-seq track. These CHiP-seq datasets were chosen after searching in the CistromDB tool. And I visualized binding patterns near gene loci using UCSC gene tracker. This supports that AR may directly regulate the genes involved in tumour progression in LGG

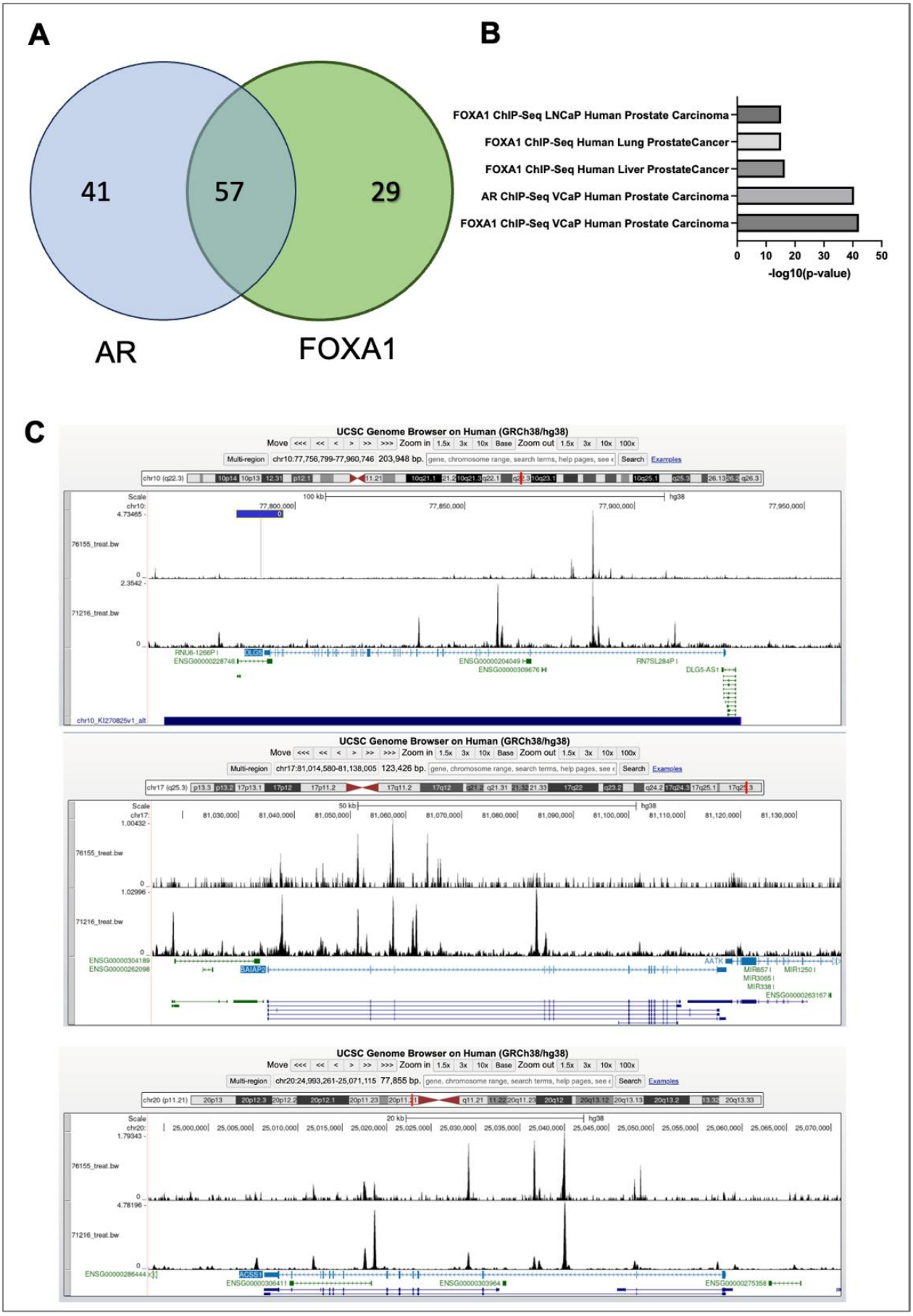


Fig 5.4: (A) The venn diagram showing the overlapping 57 genes after comparing between AR and FOXA1 ChIP-seq targets. The overlapping genes indicates co-regulation. **(B)** The bar graph showing

the functional enrichment analysis of the Transcriptional enrichment from CHEA 2022 for both AR and FOXA1 in prostate cancer cell lines. **(C)** Screenshot of the UCSC genome browser for three overlapping genes (DLG5, BAIAP2 and ACSS1) among the 57 showing the ChIP-seq binding tracks for AR and FOXA1 across the genomic loci. This suggests that both AR and FOXA1 binding near the regions of the selected genes.

6. KIDNEY RENAL CLEAR CELL CARCINOMA (KIRC)

Table 6.1: Details showing the survival associations and expression patterns of 15 Nuclear Receptors in KIRC.

Nuclear receptor	SAMPLE NUMBER	Hazard ratio	P value	Worst prognosis (OS)	Subtype expression	Stage plot F (P)
THRB	516	0.43	0.00000014	0.43 (LOW)	*m1, *m2, *m3, *m4 (down)	15.6 (9.4e-10) (down)
THRA	516	0.46	0.000001	0.46 (LOW)		15.4 (1.37e-09) (down)
RXRG	501	0.53	0.000046	0.53 (LOW)		3.92 (0.00874) (down)
RXRA	516	0.46	0.000011	0.46 (LOW)		11.1 (4.59e-07) (down)
RORC	516	0.49	0.000005	0.49 (LOW)		14.3 (5.9e-09) (down)
RORA	515	0.39	0.0000000064	0.39 (LOW)		11.5 (2.71e-07) (down)

RARB	515	0.52	0.000029	0.52 (LOW)		16.1 (5.02e-10) (down)
PPARG	516	0.46	0.00000062	0.46 (LOW)		16.9 (1.86e-10) (down)
PPARA	516	0.47	0.0000021	0.47 (LOW)	*m3, *m4 (down)	5.02 (0.00193) (down)
NR3C2	512	0.41	3e-08	0.41 (LOW)	*m1, *m2, *m3, *m4 (down)	27.3 (2.09e-16) (down)
NR2F1	516	0.57	0.00031	0.57 (LOW)	*m4 (down)	4.71 (0.00296) (down)
NR1I3	510	0.58	0.00049	0.58 (LOW)	*m1, *m2, *m3, *m4 (down)	
NR1I2	501	1.4	0.027	1.4 HIGH)		
NR1H4	516	0.68	0.014	0.68 (LOW)	*m1 (up)	
NR1D2	516	0.68	0.015	0.68 (LOW)	*m2, *m4 (down)	9.37 (4.86e-06) (down)
HNF4A	516	0.62	0.002	0.62 (LOW)		
HNF4G	516	0.49	4.4e-06	0.49 (LOW)	*m1 (up)	

ESRRG	516	0.54	0.00011	0.54 (LOW)	*m1, *m2, *m3, *m4 (down)	12.2 (1.06e-07) (down)
ESRRA	516	0.58	0.00042	0.58 (LOW)		
AR	516	0.43	1.2e-07	0.43 (LOW)	*m1 (up)	4.58 (0.00357) (down)

When it comes to the Kidney renal cell carcinoma there are many nuclear receptors to look at and I found it very interesting because 14 out of 15 Nuclear receptors **(Table 6.1)** were all down regulated across both box plot and stage plot. So instead of focusing on one receptor, I chose to look at all these 15 receptors

Table 6.2: Shows the details of Transcriptional Regukatory network of 15 Nuclear receptors associated with KIRC

	TARGETS	**TRANSCRIPTIOAL FACTORS**
NR2F1	THRB, THRA, RXRA, RXRG, RORA, RORC, RARB, PPARA, NR2F1, NR1I3, NR1D2, NR3C2, AR	THRB, **RXRA, PPARG, NR**2F1, **AR**
ESRRG	NOT FOUND	**RXRA, PPARG, AR**
AR	THRA, THRB, RXRA, RXRG, RORA, RORC, PPARA, PPARG, NR2F1, NR3C2, NR1I3, NR1D2, ESRRG, AR, RARB	**RXRA, PPARG**, NR2F1, **AR**
PPARG	RXRA, RXRG, RORA, RORC, PPARA, PPARG, NR2F1, NR3C2, NR1I3, NR1D2, ESRRG, AR, THRA, THRB, RARB	RORA, **RXRA, PPARG, AR**

PPARA	PPARA	THRB, **RXRA**, PPARA, **PPARG**, NR2F1, **AR**
NR3C2	NOT FOUND	**RXRA, PPARG**, NR2F1, **AR**
RXRA	THRA, THRB, RXRG, RXRA, RORA, RORC, RARB, PPARG, PPARA, NR2F1, NR3C2, NR1I3, NR1D2, ESRRG, AR	**RXRA, PPARG**, NR2F1, **AR**
THRA	NOT FOUND	**AR, RXRA, PPARG**, NR2F1
RXRG	NOT FOUND	PPARG, **RXRA, PPARG**, NR2F1, **AR**
NR1I3	NOT FOUND	**AR, PPARG**, THRB, **RXRA**, NR2F1
RARB	RARB,	**PPARG, AR, RXRA**, RARB, NR2F1
THRB	THRB, PPARA, NR2F1, NR1I3, NR1D2	THRB, **RXRA, PPARG**, NR2F1, **AR**
RORC	NOT FOUND	NR2F1, **RXRA, PPARG**, NR2F1, **AR**
NR1D2	NOT FOUND	THRB, **RXRA, PPARG**, NR2F1, **AR**
RORA	PPARG,	**RXRA, PPARG**, NR2F1, AR

KIRC represented very different pattern in my analysis. While other three cancers showed strong associations with two or three receptors, KIRC really stood out as 14 out of 15 receptors showed statistically significant survival patterns with HR<1 which indicates tumour suppressive roles, and it also demonstrated consistent down regulation across different tumour types and stages as shown in **Fig 6.1, Fig 6.2 and Fig 6.3**. The selected three receptors which is the AR, PPPARG and RXRA are important in this study that is explained below. Like for example AR, PPARG and RXRA all have HR between 0.41-0.46 with extremely low p-values and these receptors were also downregulated

across different types from m1 to m4. Also, while analyzing the stage plot, some receptors have very low p-values and high F value, for example PPARG, the F value is 16.9 and p value= 1.86e-10, for NR3C2 F value= 27.3 and p value= 2.09e-16. (The F-value in stage plot is derived from one-way ANOVA tests comparing gene expression across different tumour stages, or allow to identify any significant differences qcross tumour stages)

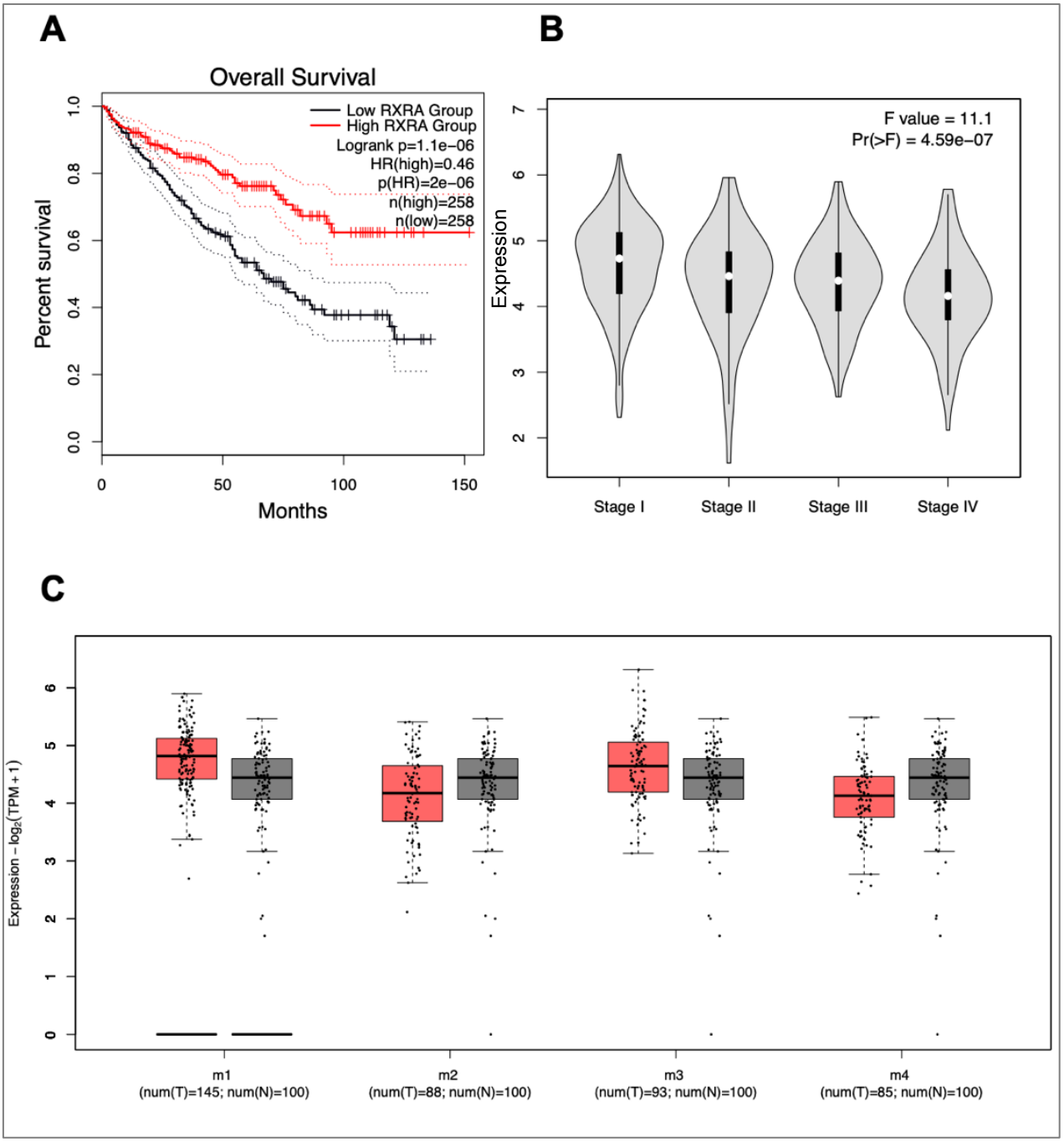


Fig 6.1: Survival significance and expression analysis of RXRA in KIRC. (A) The Kaplan-Meier survival curve which compares the overall survival between KIRC patients with high and low expressions of RXRA. The higher expression of RXRA is involved in better survival with a hazard ratio of 0.46, which indicates tumour suppressive roles in KIRC. **(B)** Shows the stage plot of RXRA expression across four different stages (I-IV). The expression of RXRA consistently downregulates

in all these stages. **(C)** The box plot of RXRA expression across all the four subtypes of KIRC. As shown in the figure, the there is no significant difference was detected.

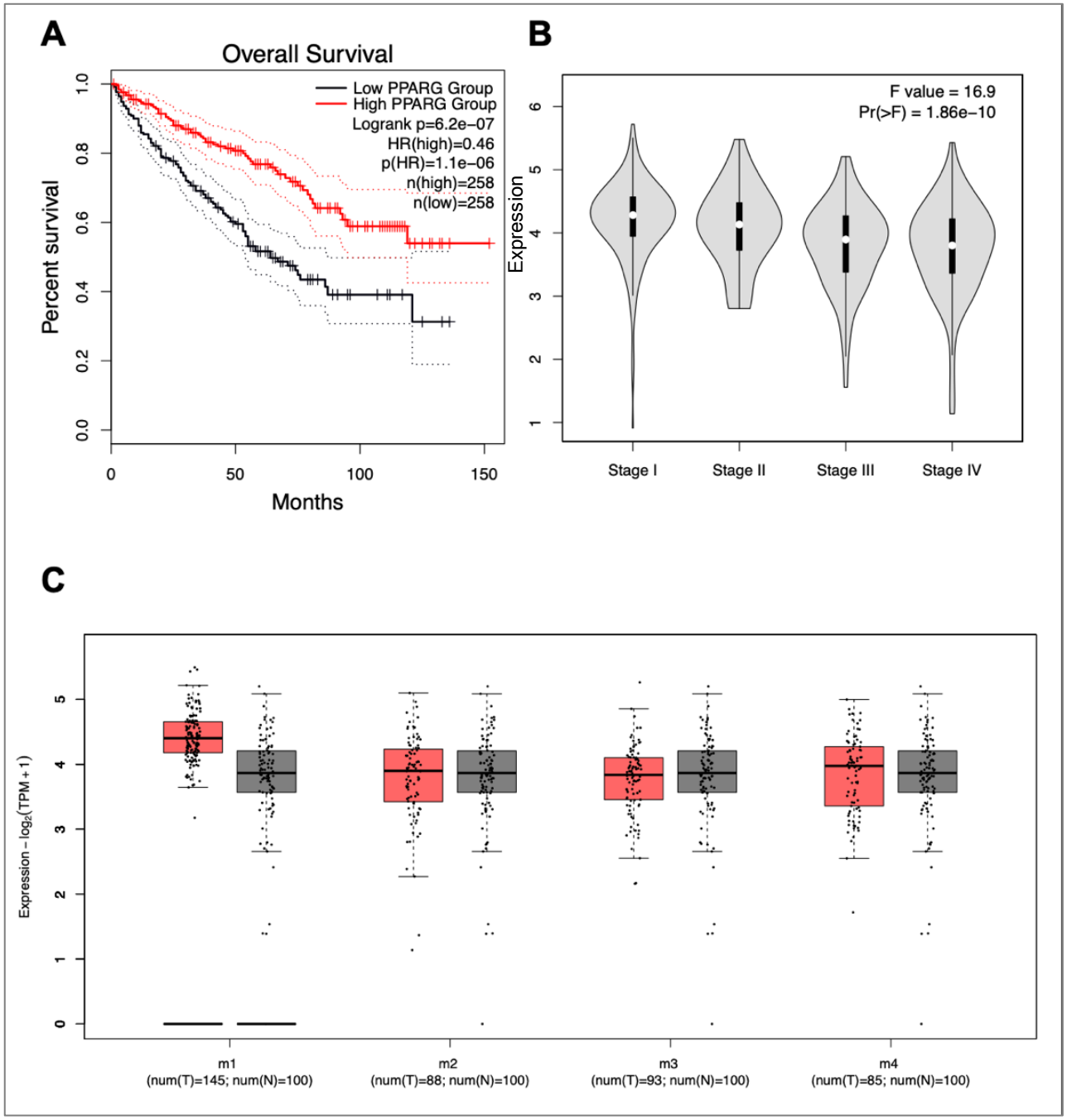


Fig 6.2: Survival significance and expression analysis of PPARG in KIRC. (A) The Kaplan-Meier survival curve which compares the overall survival between KIRC patients with high and low expressions of PPARG. The higher expression of PPARG is involved in better survival with a hazard ratio of 0.46, which indicates tumour suppressive roles in KIRC. **(B)** Shows the stage plot of PPARG expression across four different stages (I-IV). The expression of PPARG shows consistent downregulation as cancer progresses. **(C)** The box plot of PPARG expression across all the four subtypes of KIRC. As shown in the figure, the there is no significant difference was detected.

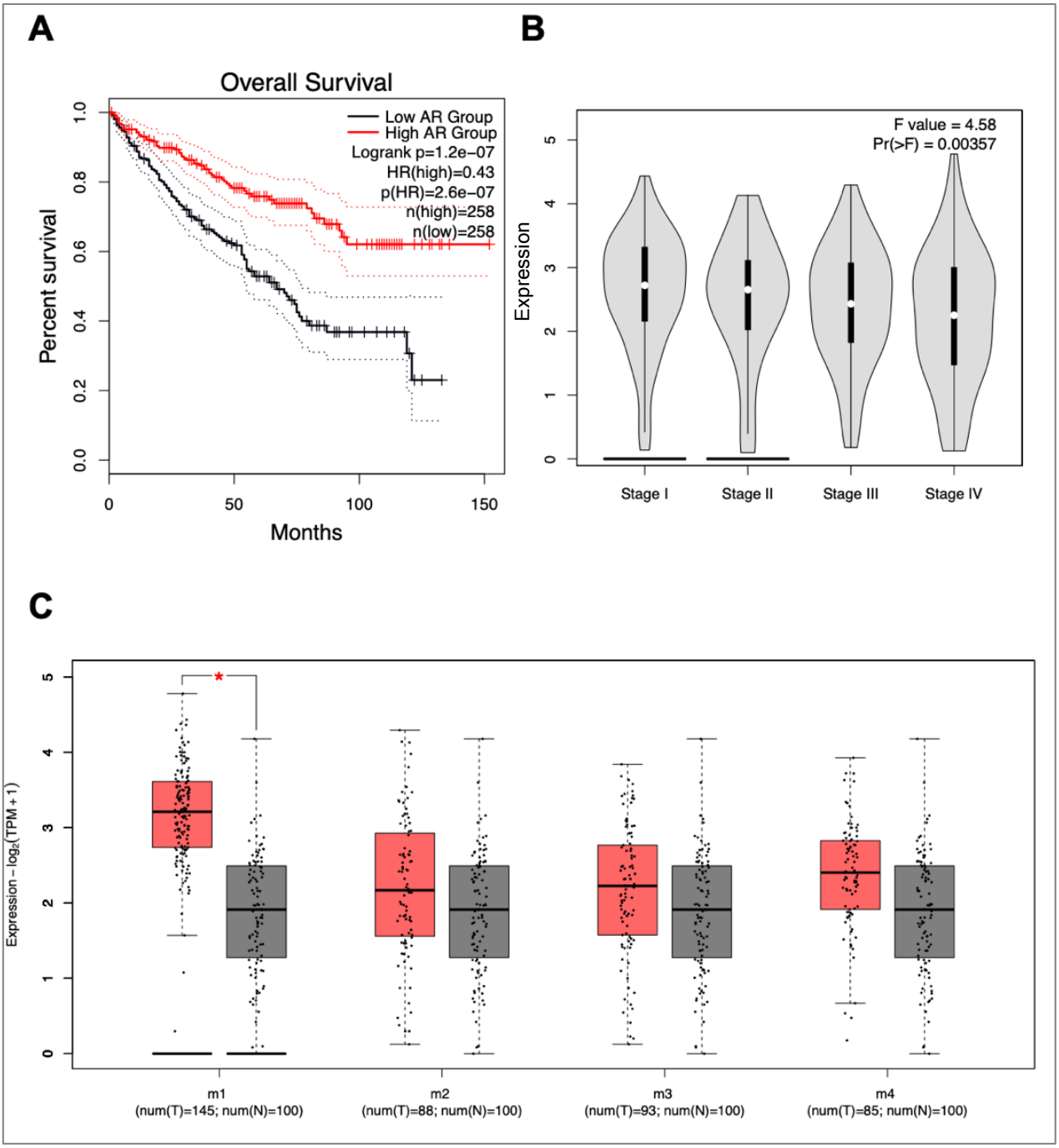


Fig 6.3 Survival significance and expression analysis of AR in KIRC. (A) The Kaplan-Meier survival curve which compares the overall survival between KIRC patients with high and low expressions of AR. The higher expression of AR is involved in better survival with a hazard ratio of 0.46, which indicates tumour suppressive roles in KIRC. **(B)** Shows the stage plot of PPARG expression across four different stages (I-IV). The expression of PPARG shows consistent downregulation as cancer progresses with F(P) value equals to 0.00357. **(C)** The box plot of PPARG expression across all the four subtypes (m1-m4) of KIRC. As shown in the figure, there is significant upregulation of AR in m1 subtype compared to the normal tissue.

All the nuclear receptors associated with KIRC are characterized by downregulation. therefore, instead of analyzing or narrowing it down to one, I did the analysis for all the receptors together.

Firstly, to understand the interrelation between these receptors, I performed the STRING network analysis, in which I visualized interaction and those capable of co regulation **(Fig 6.4).** After that I did the TF-Link analysis which allowed us to map the Transcriptional factor-target relationship between the receptors. I created a table showing which receptors act as transcription factors and which receptor are transcriptionally regulated (targets) **(Table 6.2)**.

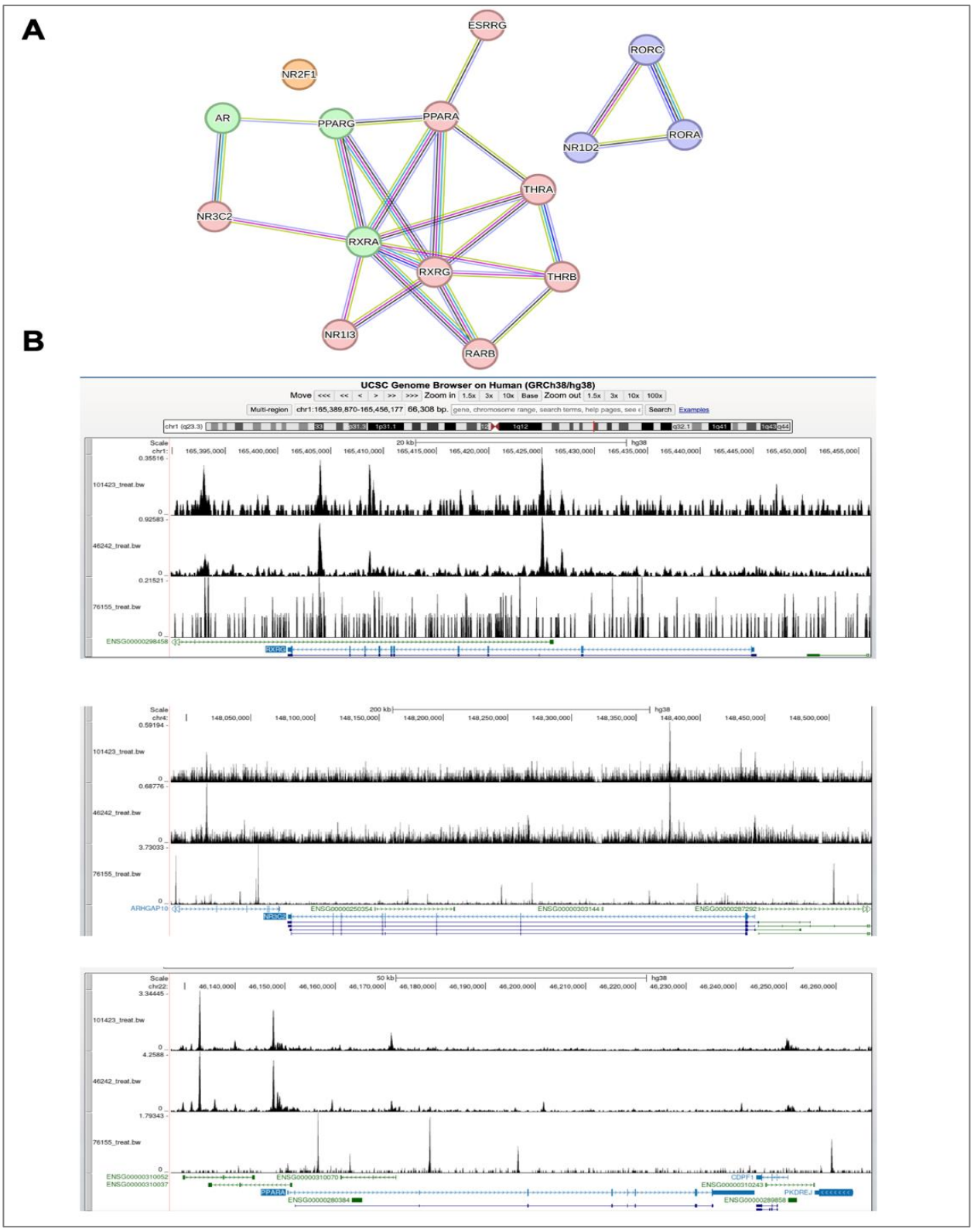


Fig 6.4: (A) The STRING interaction network depicting the functional and predicted known protein-protein interaction of all the 15 nuclear receptors associated with KIRC. Key nodes such as RXRA, AR and PPARG depicts multiple connection which directly supports their role as master regulators within that network. **(B)** UCSC genome browser for the three nuclear receptors relevant to KIRC across the three gene loci**.** The chosen receptors are RXRG, PPARA and NR3C2. These observations in the figure supports the transcriptional connections among the NRs involved in tumour suppression and progression in KIRC

From this analysis, I identified three master regulators and those are RXRA, PPARG and AR. These three transcription factors were shown to regulate the 12 other nuclear receptors and act as the central controllers or regulators. So, targeting these receptors might help to alter the cancer progression

7. LIVER HEPATOCELLULAR CARCINOMA (LIHC)

Table 7.1: Details showing the survival associations and expression patterns different nuclear receptors in LIHC

Nuclear receptor	SAMPLE NUMBER	Hazard ratio	P value	Worst Prognosis (OS)	Subtype Expression	Stage plot F (P)
PPARG	364	1.8	0.00074	1.8 (HIGH)	*icluster_3 (up)	
PPARD	361	1.5	0.028	1.5 (HIGH)		
NR2C1	364	1.7	0.0024	1.7 (HIGH)		4.8 (0.00276) (up)
NR1H3	364	1.6	0.0061	1.6 (HIGH)		
ESR1	364	0.55	0.00069	0.55 (LOW)	*icluster_1, *icluster_2, *icluster_3 (down)	3.01 (0.0304) (down)

Well, there are many potential nuclear receptors to look at, but I particularly found PPARG as the most interesting out of them. So here I the box plot and stage plot PPARG as shown in **(Fig 7.1).** this showed consistent patterns across different subtypes

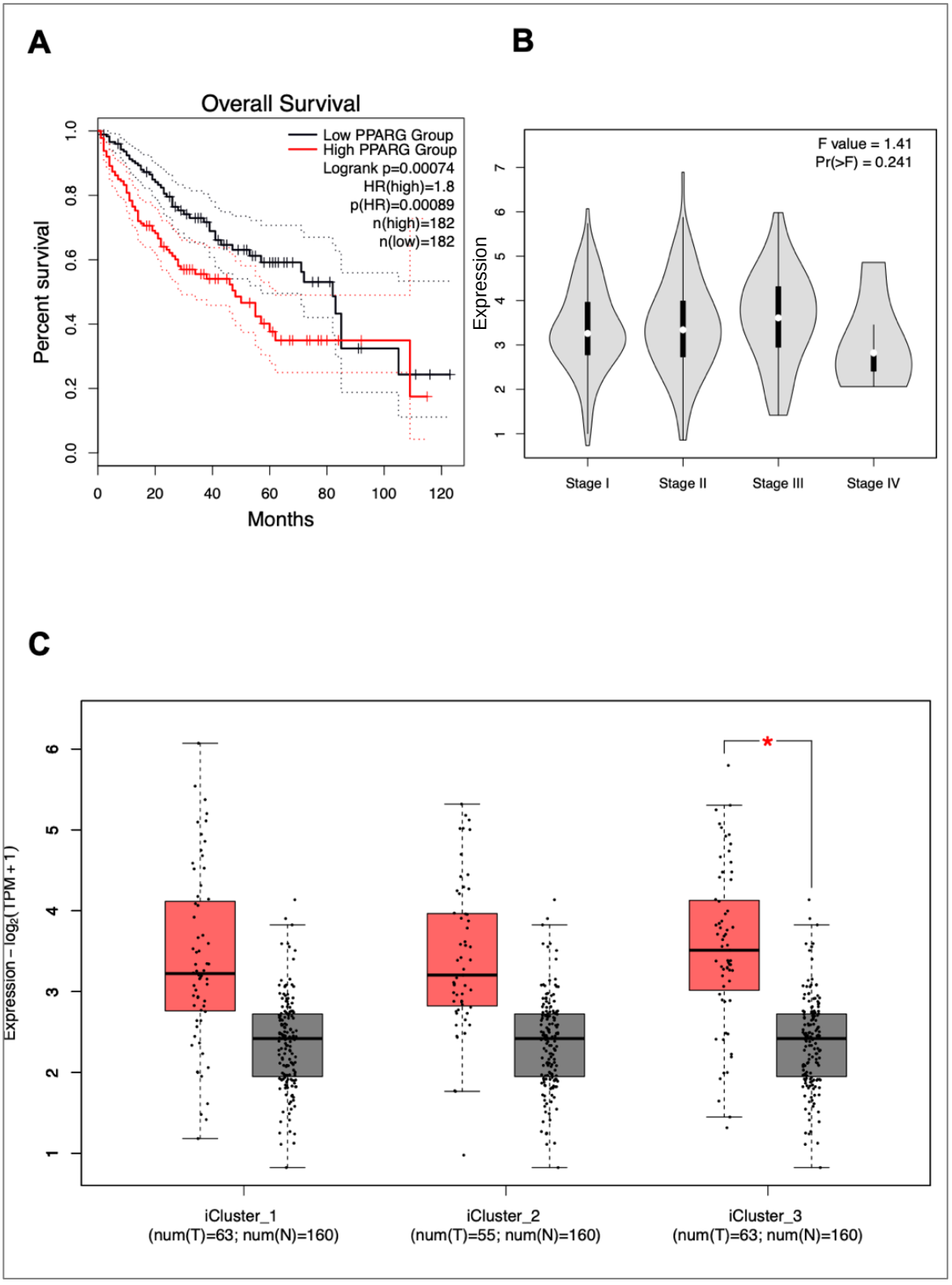


Fig 7.1: (A) Kaplan Meier survival curve that comparing the overall survival in LIHC patients with high and low expression of PPARG in LIHC. The higher expression of PPARG in LIHC is involved in worst prognosis with a hazard ratio= 1.8 and depicts a significant p value <0.05. This indicates that they are involved in oncogenic roles in LIHC. **(B)** The stage plot showing the expression of PPARG in all the four stages. There is no statistical significance were observed from the plot as the F(P) value is 0.241. **(C)** The box plot demonstrating the expression of PPARG in all the three subtypes of

liver cancer. In the figure the expression of PPARG is significantly higher in iCluster_3 subtype than the normal tissue.

So, in LIHC several NRs showed significant survival patterns with HR>1.5 which indicates potential oncogenic roles. Among these receptors PPARG (Peroxisome Proliferator-Activated Receptor Gamma) emerged as good candidate for further analysis. While analysing I found that it had a Hazard ratio of 1.8 with a p-value of 0.00074 **(Table 7.1)**, making it as an indicator of worst prognosis in this type of cancer. Also, the expression of PPARG was increase in the iCluster_3 subtype. This upregulation along with the increased hazard ratio might suggest that PPARG play a crucial oncogenic role in LIHC.

Although other receptors like the PPARD and NR2C1 also showed increased hazard ratio i.e 1.5 and 1.7 respectively, they both had weak p-values and less consistent patterns across subtypes and different stages. But LIHC showed a stronger statistical significance compared to others.

So based on these finding, I wanted to know how PPARG significantly contributes to LIHC, particularly in context of treatment resistance. Different studies and literature review proved that PPARG is involved in regulating lipid metabolism, which is important in hepatocellular Carcinoma. I used this as a starting point to connect the resistance involved in different treatments.

To test this, I first performed two analyses, **1)** I analyzed the gene expression data from a sorafenib-resistant hepatocellular carcinoma cell lines referred to as resistant clone. A comparison was done with Resistant Clone (368 genes) vs Sorafenib regulated genes. The gene list was obtained from a dataset in GEO2R. In this I started by comparing a sorafenib-resistant LIHC cell lines with genes that are regulated by sorafenib, a first line treatment for advanced liver hepatocellular carcinoma. This comparison helped in the identification of genes that are contributing to drug resistance in LIHC, i.e. altered under sorafenib treatment and resistant clone So, when I compared these two, I got 156 overlapping genes as shown in **Fig 7.2 (A)**. This was done to show those genes that are commonly affected or expressed in both groups under sorafenib treatment. **2)** Resistant genes to PPAR ligands ie troglitazone and pioglitazone. These gene list from was also downloaded from GEO2R under a dataset. While comparing these ligands with resistant genes, the Troglitazone showed a very good overlap, which indicates that it may reverse resistance by altering the PPARG activity **(Fig 7.3)**.

As shown in Fig 15 troglitazone (TROG) showed a larger number of genes and 98 was found to be overlapping between resistant clones and TROG. So, I did the functional enrichment analysis using ENRICHR with the 98 genes along with 9 common overlapping genes found in all the three sets to see what they regulate. I found that they regulate lipid metabolism, growth signaling pathways and PPAR signaling pathways which demonstrates the central and crucial role of PPARG.

Out of curiosity I wanted to know what those resistant clones (368 genes) regulate. So, I did the functional enrichment analysis using ENRICHR and found that they regulate lipid metabolism, growth signaling pathways. To represent that in a graphical manner, the bar graph showing the biological processes that is being regulated by these genes were shown in **Fig 7.2 (B).** And the interesting thing I found is that under the KEGG 2021 HUMAN, the PPAR signaling pathway was noted which

indicates that these genes also regulate the PPAR signaling.

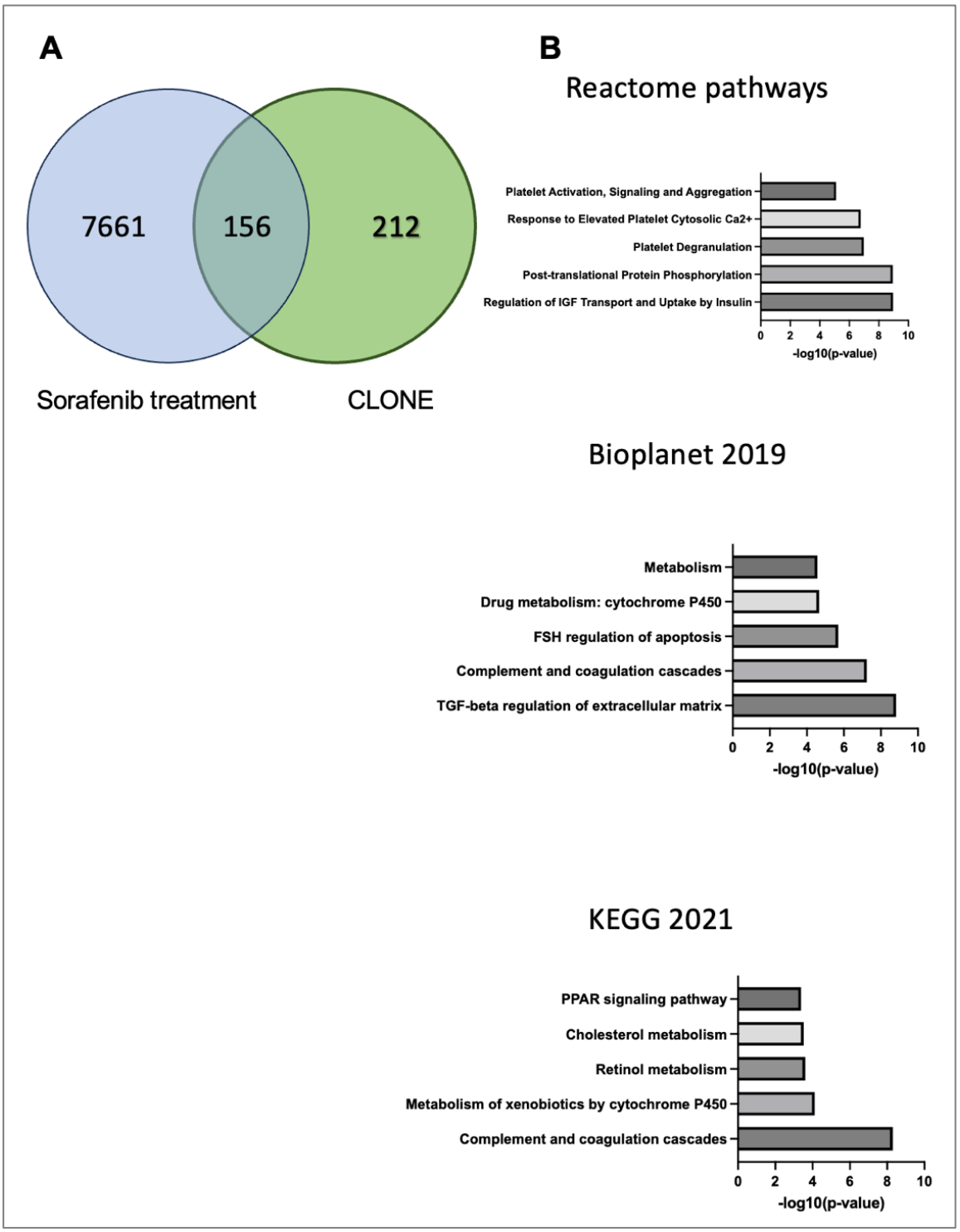


Fig 7.2: (A) The venn diagram showing the intersection between the genes regulated by sorafenib treatment and differentially expressed genes the resistant clone list. A total of 156 overlapping genes were regulated by both, which indicates that they are involved in sorafenib resistance mechanism. **(B)** The functional enrichment analysis of the 156 overlapping genes were done using the Reactoime pathways, Bioplanet 2019 and KEGG 2021Human databases. Observed PPARG signaling being enriched which suggests its involvement in drug metabolism and lipid metabolism in Liver cancer

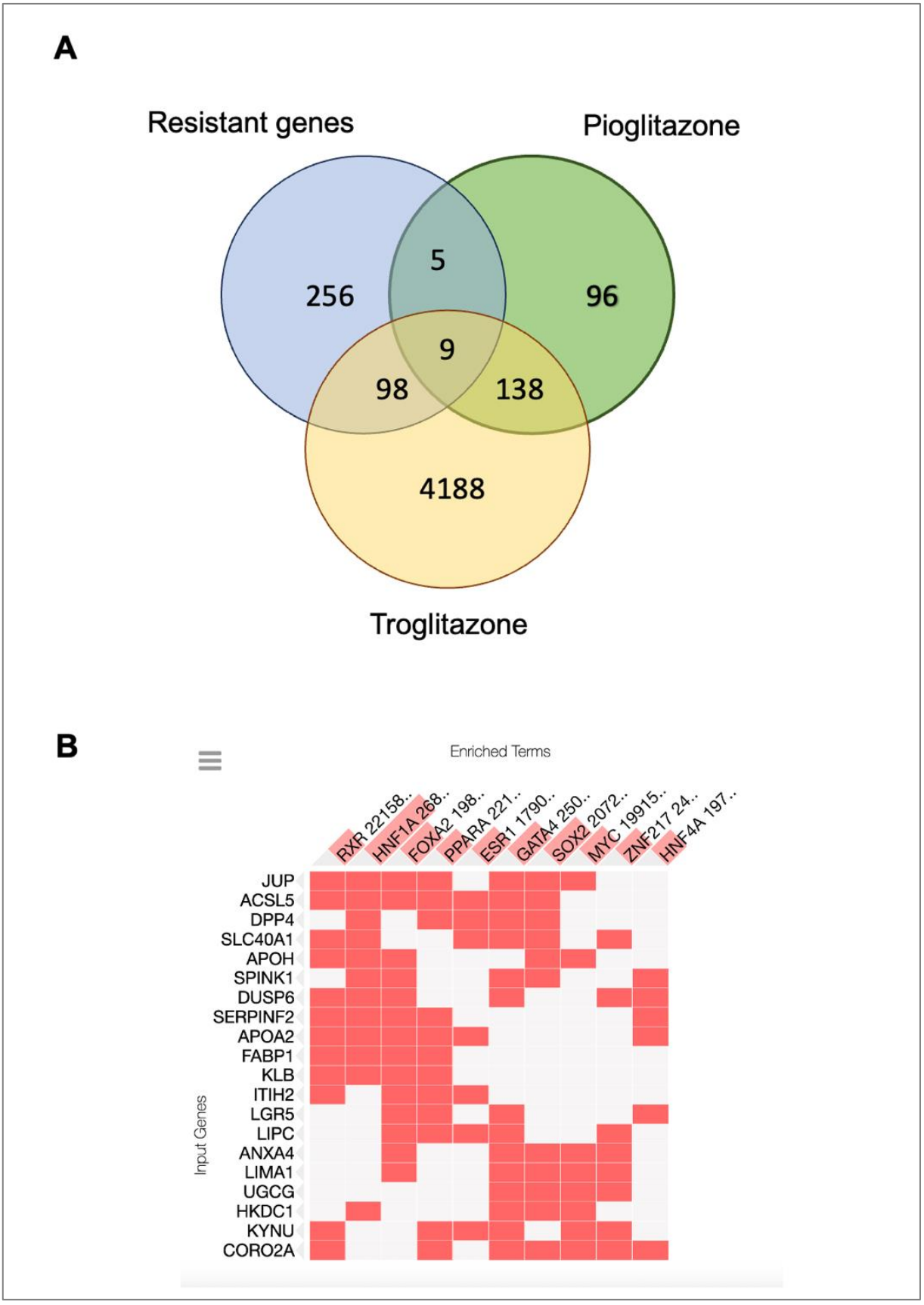


Fig 7.3: (A) The Venn diagram showing the overlapping genes between the sorafenib resistant clone and the genes regulated by the PPAR ligands Troglitazone and Pioglitazone. A total of nine genes were commonly regulated by all the three and the TROGLITAZONE showing a greater overlap with the resistant (98 genes) compared to Pioglitazone. This demonstrates reverse resistance via the

modulation of PPARG. **(B)** An enrichment analysis was done for the 98 genes associated with Troglitazone and resistant genes and the 9 overlapping genes making a total of 107 genes. This heat map demonstrates the important transcription factor that regulates the input genes shown in the y-axis. This indicates the involvement of NRs in resistance.

To know the clinical importance, I compared the Top 500 LIHC with the resistant clones, which surprisingly gave me 4 overlapping genes and those are E2F8, SCML2, SLC9A3R1 and SOCS2**(Fig 7.4 (A))**. Among them I found that E2F8 is a transcription factor, so the next thing I did was to search E2F8 in CistromeDB to identify its putative targets. I also downloaded the putative targets from CistromeDB to know its transcriptional coordination. I got the gene list, and I compared these with the Top 500 LIHC as shown in **Fig 7.4 (B).**

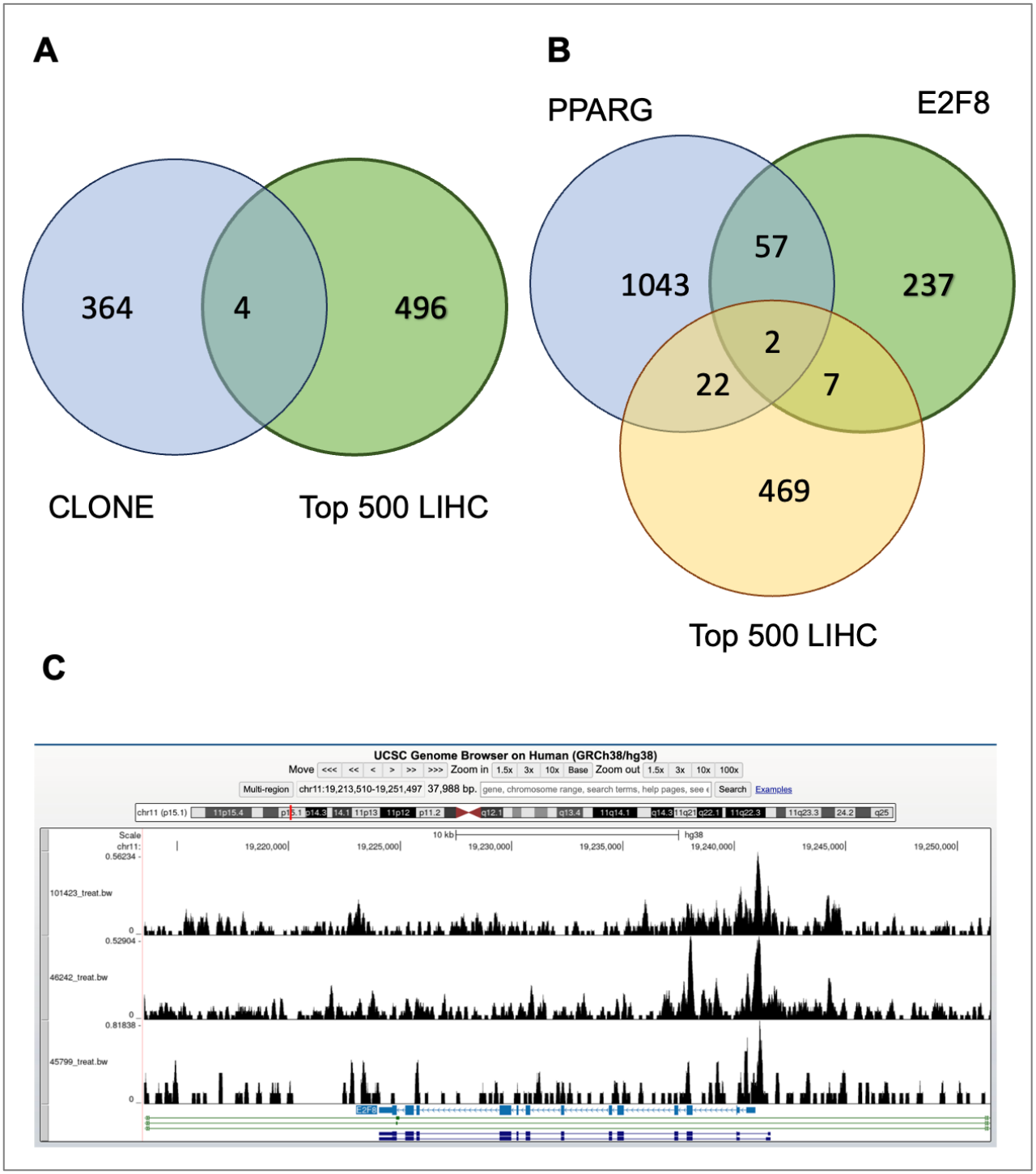


Fig 7.4: The Venn diagram showing the 4 overlapping genes after comparing the resistant clone genes with Top 500 survival associated genes in LIHC. Among the 4 genes E2F8 transcription factor was observed. **(B)** A venn diagram showing the E2F8 and PPARG ChIP-seq with the Top 500 survival associated genes in LIHC. This indicates very high clinical relevance. **(C)** The UCSC gene browser track illustrating the the signal peaks at E2F8 gene locus which confirms the PPARG and E2F8 binding, this directly states that it is involved in drug resistance and plays an important role in liver cancer progression.

All this performed analysis again proves that PPARG is not only involved poor and worst prognosis in LIHC but also showed that it may be directly driving drug resistance in liver mainly towards sorafenib treatment. However, the use of Troglitazone can reverse this effect i.e. it is involved in reverse resistance.

8. DISCUSSION

This study used in-silico bioinformatic analysis to specifically identify nuclear receptors with clinical and regulatory importance across four different cancer types, ACC, LGG, KIRC and LIHC. The key findings of this study include the potential tumour suppressive role of THRB in ACC, the oncogenic role of AR in LGG, the transcriptional network among three master regulators AR, PPARG and RXRA in KIRC and the involvement of PPARG in drug resistance in LIHC. These results were produced by analyzing only four types of cancer and supports the involvement of nuclear receptors in tumour progression.

8.1. THRB in Adrenocortical Carcinoma (ACC)

In Adrenocortical carcinoma (ACC), THRB expression plays a protective role as low expression is associated with tumour progression. This is done by regulating the genes involved in lipid metabolism. ACC is an aggressive tumour with poor survival outcomes (31). As we know the thyroid hormone receptor THRB is a key regulator of cellular growth and differentiation that is being dysregulated in various cancer. The main function of THRB is to regulate the gene expression in response to its ligand T3 (triiodothyronine). Earlier studies have proven the involvement of THRB also called TRβ in suppressing the PI3K signaling in anaplastic thyroid cancer. THRB is shown to be a tumour suppressor in different types of cancer like breast cancer, thyroid cancer, hepatocellular carcinoma and other solid tumours. Certain other studies have shown that the decreased expression of THRB leads the cancer to progress aggressively. However, restoring its expression in cancer cell lines can reduce its aggressiveness (32,33). Experimental approaches have shown that THRB suppresses the tumour growth in primary breast cancer and metastasis (34). All this evidence strongly suggests that THRB plays a tumour suppressive role in various cancers.

In this in-silico analysis THRB is downregulated in ACC and provide the evidence of its protective role in patient survival. Previous studies also support its tumour suppressive role in different cancers by upregulating its expression.

As for now there is limited literature to support the role of THRB in ACC. But this analysis paved a way for future research of how THRB plays a protective role in ACC.

8.2 AR in Low-Grade Gliomas (LGG)

In this study, the Kaplan-Meier survival plot showed that the increased expression of Androgen receptor (AR)is involved in poor survival in LGG. The box plot also indicates the higher expression of AR when compared to the normal brain tissues. The genes regulated by AR ligands such as the R1881, Testosterone and DHT were also found to be enriched in FOXO signaling and Rho-GTPase signaling pathways. Another finding observed in this study is the co-regulation of target genes by AR and FOXA1. Various experimental studies have shown the increased expression of AR leads to the aggressive LGG and that is stated below.

AR shows significantly increased expression in gliomas. Various studies have stated that AR protein is more abundant in glioblastoma than in LGG. In LGG the overexpression of AR is associated with worst overall survival (OS) (5-years) (35). A pan cancer analysis study of AR in various cancer types resulted in decreased expression of AR in 12 cancer types while glioblastoma multiforme showed increased expression. The survival analysis showed that high AR mRNA expression is associated with poor survival (36). However, experimental approaches proved that an inhibitor called enzalutamide suppresses the expression of FOXA1 that in turn caused the downregulation of AR (37). The AR-FOXA1 coregulation is studied mainly in prostate cancer, several literatures have been found based on that. This study showed the evidence of how FOXA1 controls the activity of AR in prostate cancer. The expression analysis demonstrated that 41 % of androgen receptor variants were co-regulated by FOXA1 with good overlap. In this study they also compared the genes regulated by AR with a different study conducted by Lu.et al, 2015 and found set of genes common in both studies. 50% of the targets were found to be regulated by FOXA1 (38,39).

This suggests that FOXA1 is helping AR in controlling the gene expression which proves that they are a key co-regulator.

8.3 AR, PPARG and RXRA in KIRC

In Kidney renal clear cell carcinoma (KIRC), 15 nuclear were shown to be downregulated during this in-silico analysis. The obtained transcriptional network using TF-Link suggested that the nuclear receptors AR, PPARG and RXRA may act as master regulators that control the expression of other 12 nuclear receptors. Higher expression of these receptors was involved better overall survival.

The Kidney renal clear cell carcinoma (KIRC) is most common subtype of RCC (Renal cell carcinoma). A research study shown that the low HR value PPARG plays a potential protective role in KIRC. A shown in **Fig 6.2 (A),** the higher expression of PPAR is associated with better survival. This experimental approach proved that this model predicts 5 and 10-year survival. The lower expression of PPARG is associated with aggressive tumour and worst prognosis. And noted a lower expression of PPARG in patients with metastasis (40,41). Another study conducted an experiment by deleting the PPARG in Clear cell renal cell carcinoma (ccRCC), where it showed no difference in

viability, proliferation or tumour growth in mice. Although the in-silico analysis showed that the increased expression of PPARG is associated with better survival in KIRC, but certain studies proved how PPARG contributes to tumour maintenance. Its states it may be stage specific i.e. PPARG is involved in tumour initiation but not progression (42).

The downregulation of RXRG in KIRC is associated with aggressive tumour growth (large size). This finding suggests that the low expression of RXRG is associated with poor survival. And my in-silico analysis predicted that the higher expression of RXRA is associated with better prognosis. This may suggest that both receptors in Retinoid X Receptors family act as a tumour suppressor in KIRC (43). However, there are a limited number of articles to support the statement that RXRA act as a tumour suppressor but most of the literatures support that RXRG is a tumour suppressor which is also a member of the RXR family this may in turn proves that RXRA is involved in repressing the tumour.

Several studies shown that the decreased expression of AR is involved in worst survival. It is important to know that the higher expression of AR in both transcript and protein levels are associated with better overall survival (44). This with this study of how higher expression of these receptors are involved in good prognosis.

The PPARG act as a master regulator for adipocyte differentiation and found to regulate the expression of most of the genes involved in lipid homeostasis. Several other studies suggested that the RXRs don't work alone but in fact act as a master regulator of various pathways (45,46).

8.4 PPARG in LIHC

In this in-silico analysis it was found that the higher expression of PPARG is associated with worst prognosis and poor overall survival. It was observed that there is a significant increased expression of PPARG in LIHC when compared to the normal tissues. PPARG is also driving resistance especially to sorafenib treatment. However, Troglitazone can reverse this resistance.

Furthermore, various studies have shown how the higher expression of PPARG in various cancers like LGG, LIHC, GBM is associated with worst survival. These studies indicate the usage of PPARG as cancer biomarker (47). The downregulation of PPARG induces cell proliferation that results in tumorigenesis (48,49).

Sorafenib is kinase inhibitor that has been used to target hepatocellular carcinoma. Studies have found that only 30% of liver cancer patients were benefitted from sorafenib due increased resistance. This resistance is driven by PPARG (50).

There is no evidence of how Troglitazone is involved in reverse resistance.

Although this study focused on ACC, LGG, KIRC and LIHC, nuclear receptors are also known to play major roles in certain cancers like breast, prostate, pancreatic, glioblastoma and stomach cancers. So at first all these were studies to pick the best out of all the cancers

Androgens play an important role in prostate cancer pathogenesis. However, other NRs such as the Vitamin D receptor, estrogen receptor β and mineralocorticoid receptor also play crucial

antioncogenic roles while peroxisome proliferator-activated receptor γ, androgen receptor, and oestrogen receptor α promote oncogenesis. Recent studies have proven the involvement of orphan receptors in the promotion of prostate cancer. Orphan receptors such as NRs, RORγ, TR2, TR4, COUP-IFII, ERRα, DAX1 and SHP interfere with androgen receptors (51,52). However, monoclonal antibodies such as bevacizumab can target VEGF as they promote angiogenesis. The chimeric antigen receptor T-cell therapy was found to be an advanced therapeutic agent against prostate cancer (53).

Pancreatic adenocarcinoma (PAAD) is a life-threatening type of cancer with a 5% survival. There is certain proof that orphan receptors and other ligand-activated NRs are involved in pancreatic cancer. The NRs are very important in the development and homeostasis of the pancreas. The main nuclear receptors known to be involved in PAAD are the retinoid receptors, peroxisome proliferator activated receptor, androgen and oestrogen receptors (14). The involvement of peroxisome proliferator- activated receptors in pancreatic cancer. The androgen receptors (AR) are mainly involved in pancreatic cancer. Some studies have shown that AR can regulate pancreatic cancer growth in vitro, and it was found that patients affected with PAAD are found to have low levels of testosterone (54). Chemotherapy is considered the basic approach to treat PAAD. However, Gemcitabine in combination with chemotherapy such as the 5-fluorouracil, capecitabine and taxane failed to show improved survival rates, but new combinations with Folfirinox and Abraxane have been discovered. This resulted in improved survival rates (55).

Some nuclear receptors like TLX are a type of orphan receptor that is highly expressed in Glioblastoma Multiforme (GBM). The interaction between TLX and transforming growth factor β (TGF-β) **(Fig 8.1)** is very crucial for regulating the cell proliferation and cancer-inducing properties of the glioblastoma cell lines (56).

The overexpression of TLX results in continuous self-renewal of Neuronal stem cells (NSCs) in the brain, which leads to the development of glioma and contributes to immune suppression by activating PD-L1 and binding to PD-1. This could be a potential therapeutic target for treating glioma (57). Three molecules through computer-aided discovery showed 40-50% inhibition towards the TLX receptor. These molecules were named VPC-33010, VPC-33017 and VPC-33035 (58).

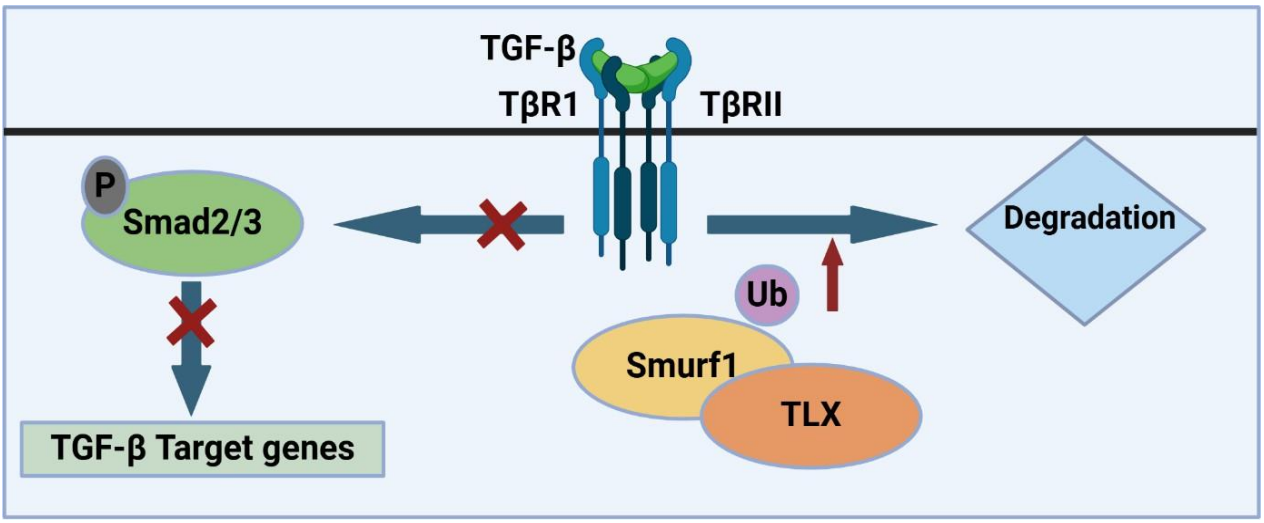


Fig 8.1: The diagrammatic representation of the involvement of TLX in TGF-β signalling inhibition in Glioblastoma Multiforme. At first, TLX interacts with E3 ubiquitin ligase smurf1 for the promotion of ubiquitination and is involved in the degradation of the TGF-β receptor type, resulting in the suppression of Smad2/3 phosphorylation, which leads to the inhibition of the TGF-β target genes. This will help in the progression of GBM that alters tumour cell proliferation, invasion and migration. (Adapted from (21))

The renal NRs mainly include the peroxisome proliferator-activated receptors, oestrogen-related receptor (ERR), Vitamin D receptors, Liver X receptors, and farnesoid X receptors. These receptors are known to regulate the progression of certain diseases (59). The nuclear receptor subfamily NR1H4 was expressed in different cancers. Recent studies have found that FXR was highly expressed in stage I clear cell renal cell carcinoma (ccRCC) and inhibiting or knocking down the FXR receptor has proven to suppress this cancer (60).

STAD or gastric cancer have caused 782,685 deaths over the past few decades. The cir-NR1P1 accelerates the progression of tumours very aggressively in stomach adenocarcinoma. With the utilization of bioinformatics techniques and tools, researchers have found the function and role of NR1P1 in STAD (61). All the members of the nuclear receptor family are somehow involved in the progression or suppression of different types of cancers. They play a crucial role in the tumour microenvironment (TME) by controlling and regulating inflammation and regulating immune responses (62).

LIMITATIONS AND FUTURE WORK

Limitations:

Despite the strength of this study, certain limitations must be acknowledged. In some instances, finding a good quality dataset is quite challenging. It was also challenging to find the treatments in the tissue type of interest. Another limitation was the difficulty in finding the relevant and appropriate ChIP-seq data of the respective tissues. Additionally, this project involved comparing the data of separate experiments done by different research groups, done at different times. These experiments also vary in the methodology and design. This can affect the efficiency in comparability.

Future work:

This study used bioinformatics methods and tools to predict the function of nuclear receptors and its relationship with different cancers. The next phase of this study will be experimentally testing these results and predictions in the lab

THRB (ACC): One of the key findings is that THRB plays a tumour suppressive role in ACC by regulating the genes involved in lipid metabolism. To validate this finding treat any ACC cell lines with thyroid hormone and do the PCR for any genes that is found in the overlap involved in lipid metabolism. This will help in confirming whether THRB is involved in upregulating or suppressing these genes which supports its predicted role in ACC.

AR (LGG): Here, AR was found to be involved in worst and poor prognosis and predicted that it regulates FOXO signalling. This can be validated by treating the LGG cell lines with AR agonists like the R1881 or DHT or Testosterone and perform PCR on any of the AR target genes maybe that is involved in FOXO signalling. This will in turn confirm the regulatory role of AR in LGG.

AR, PPARG and RXRA (KIRC): The in-silico findings proved that AR, PPARG and RXRA act as master regulators among the group of 12 nuclear receptors involved in KIRC. To validate this, knock down the expression of AR, PPARG and PPARG. Then the PCR can be done to look at the expression of the other 12 nuclear receptors to check their expression. Through this experiment we can predict that losing expression of one or more of the three master regulators leads to the downregulation of the other receptors.

PPARG (LIHC): The in-silico analysis found that PPARG is involved in drug resistance especially to sorafenib treatment and predicted that Troglitazone is involved in reverse resistance that it can reverse this effect. We can validate this by taking the liver cancer cells i.e. HepG2 and making them sorafenib resistant. Then treat with or without Troglitazone. After that PCR can be done to measure the expression of a few targets that is identified in the in-silico studies say E2F8. This will help in confirming whether Troglitazone is involved in modifying the expression of the genes that drives resistance. This maybe validate the role of PPARG in LIHC treatment.

CONCLUSION

This in-silico bioinformatics study explored the clinical and regulatory roles of nuclear receptors across 4 different cancer types i.e. ACC, LGG, KIRC and LIHC. Key findings predicted from this study involves THRB as tumour repressor in ACC, AR as a oncogenic regulator in LGG, AR/PPARG and RXRA as master regulators for other receptors in KIRC and PPARG as a driver of drug resistance towards sorafenib treatment. These findings will pave a basement of future in-lab experiments.

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